

Machine-Learning-Assisted Empirical Bayes Estimation for Regression on Grouped Network Data

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Abstract

We estimate group-specific heterogeneity in two-way effects models for matched data represented by a bipartite network. When the network is sparse or connected across groups, ordinary least squares (OLS) estimates individual effects imprecisely. We propose a machine-learning-assisted empirical Bayes (EB) estimator that uses the network to estimate group memberships and then shrinks effects toward their group-specific prior distributions. We represent the model's conditional dependence structure as a Bayesian network, providing a unified framework for the two-step estimation procedure. Under stated conditions, we establish identification and consistency, and show that the feasible EB estimator approaches the oracle posterior mean asymptotically, even when a vanishing fraction of nodes is misclassified. Monte Carlo simulations show that the proposed estimator outperforms OLS and pooled EB when effect distributions differ across groups and the network provides information about group membership. When the true distributions coincide across groups, the estimated group-specific priors converge to the common prior under our consistency conditions. Finally, in an application to Amazon user-product ratings, we estimate and characterize heterogeneity in users' rating tendencies and product-specific rating components.

Keywords: Two-way effect, empirical Bayes method, directed acyclic graph, Bayesian network, community detection, bipartite graph, matched data, shrinkage estimation.

JEL Codes: C11, C13, C18, C23, C55

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1 Introduction

Two-way effects regressions have long been a workhorse for estimating high-dimensional heterogeneity in settings involving interactions or matched data. This literature dates back to the seminal contribution of [Abowd et al. \(1999\)](#). In worker–firm data, these models separate worker-specific earnings components from firm wage premia ([Card et al., 2013](#)). In student–teacher data, they distinguish student effects from teacher contributions ([Jackson et al., 2014](#)). Other important applications include estimating firm and bank effects ([Amiti and Weinstein, 2018](#)), patient and healthcare-location effects ([Finkelstein et al., 2016](#)), and family and neighborhood effects ([Chetty and Hendren, 2018](#)), among many others. More recently, researchers have linked two-way effects models to network analysis by representing matched observations as a bipartite network. [Jochmans and Weidner \(2019\)](#) show that the precision of least-squares estimates depends on the network’s connectivity. In empirical applications, weak connectivity, which may arise in sparse networks, can make individual effects difficult to estimate precisely and is associated with the limited-mobility problem. To address this challenge, [Cheng et al. \(2025\)](#) develop an empirical Bayes estimator that uses assortative matching patterns to inform the prior distribution and shrink the estimated effects. Unlike conventional EB estimators, their approach exploits dependence structured by the network, using observed links to relate effects on the two sides.

Yet the distribution of individual effects may also vary across groups. Previous work documents grouped heterogeneity in settings such as cross-country outcomes and worker productivity ([Bonhomme and Manresa, 2015](#); [Bonhomme et al., 2022](#); [Okui and Wang, 2021](#)). Therefore, a common prior may pool individuals whose effects follow different distributions. In this paper, we allow the distributions of heterogeneity to vary across groups on both sides. Our method first estimates these groups by machine learning from the observed bipartite network and then uses the estimated group-specific distributions for empirical Bayes shrinkage. This approach borrows information across nodes with similar matching patterns while preserving heterogeneity across groups. Whereas [Cheng et al. \(2025\)](#) use normalized matching frequencies to induce cross-side dependence in the prior, our method uses the network to estimate latent groups and allows the marginal distributions of effects to vary across those groups.

By assuming a group structure in two-way heterogeneity, we develop an empirical Bayes estimator that uses latent groups in a bipartite network to guide how information is pooled across

individuals. The central idea is that matching patterns can help identify a suitable distribution toward which an imprecisely estimated effect should be shrunk. When units belong to distinct types and links represent observed interactions, matching patterns may contain information about the distributions of the effects themselves. For example, workers with similar productivity may sort into similar firms ([Bonhomme et al., 2019](#)).

We design a Bayesian network that summarizes the model’s conditional dependence assumptions. The network encodes the assumed relationships among observed outcomes, network links, latent effects, group memberships, and prior hyperparameters. This representation clarifies the data-generating process and provides a unified framework for estimation by integrating group detection with the specification of the prior and likelihood. Under this dependence structure, the joint distribution factorizes into a product of conditional distributions. Once group memberships are estimated and treated as fixed, the posterior distribution of the effects is proportional to the likelihood times the group-specific prior. We estimate the hyperparameters from the data and plug them into this posterior to obtain the empirical Bayes estimator, which shrinks imprecisely estimated effects toward their estimated group-specific distributions.

Our method uses a two-step estimation procedure. First, it identifies the group structure on each side of the network. Second, it estimates a separate prior distribution for each group. The resulting posterior mean combines each unit’s observed outcomes with the prior distribution associated with its estimated group. Estimation error therefore has two sources: group misclassification and estimation of the prior hyperparameters. Under conditions that control the misclassification rate and ensure consistent estimation of the hyperparameters, the proposed estimator performs well. When the true effect distributions are common across groups, the estimated group-specific prior parameters asymptotically converge to the same population values under the consistency conditions. In finite samples, however, estimating these priors separately can add a small amount of mean squared error relative to pooled empirical Bayes, as the negative-control simulations illustrate.

Our theory clarifies what the outcome data identify and what the prior contributes. Each observed outcome contains the sum of one effect from each side of the network, so adding a constant to all effects on one side and subtracting it from all effects on the other leaves the fitted outcomes unchanged. We resolve this location ambiguity by setting the weighted average of the product-side group prior means to zero; this normalization does not constrain the realized effects. Under the stated Gaussian model, network-recovery conditions, and likelihood

conditions, the group-specific prior parameters are consistently estimated. The feasible empirical Bayes estimator then asymptotically approaches, in average squared distance per node, an oracle estimator that knows the true groups and prior parameters. This result allows a vanishing fraction of group labels to be misclassified; it does not require precise estimation of every individual effect. The theory treats the group counts as fixed, while the paper evaluates group-count selection separately.

The Monte Carlo study varies the network’s average degree, the strength of assortative matching by group, and the extent of between-group differences in the means, variances, or both of the effect distributions. We compare the proposed machine-learning-assisted empirical Bayes (EB) estimator with two-way least squares of [Jochmans and Weidner \(2019\)](#), pooled EB, and the specialized EB estimator of [Cheng et al. \(2025\)](#). We also consider infeasible benchmarks that use the true group memberships or prior parameters. In the main design, the proposed estimator achieves lower mean squared error (MSE) than the feasible benchmarks. Its gains are largest when effect distributions differ across groups and the network provides sufficient information to recover group membership. When the group-specific distributions are identical, estimating separate priors leads to a small increase in MSE relative to pooled EB. This penalty is negligible compared with the gains in designs with pronounced distributional heterogeneity and a stronger matching pattern.

The empirical application examines user–product ratings in Amazon’s Industrial and Scientific category. The 5-core, de-duplicated sample contains 412,947 ratings from 50,985 users on 25,848 products. The network-clustering procedure produces 10 user groups and 11 product groups, with differences across groups in both estimated prior means and dispersions. These estimates describe persistent user-specific rating tendencies and product-specific rating components conditional on the additive model. A supplementary held-out-rating exercise finds that proposed EB shrinkage improves prediction relative to OLS estimator.

This paper connects several strands of research. First, it builds on econometric work that allows parameters to vary across latent groups; examples include [Su et al. \(2016\)](#), [Chetverikov and Manresa \(2022\)](#), [Guðmundsson and Brownlees \(2021\)](#), and [Chen et al. \(2023\)](#). Second, it relates to the empirical Bayes and shrinkage literature, which pools information to improve the estimation of high-dimensional heterogeneous parameters ([Robbins, 1992](#); [Efron, 2012](#); [Ignatiadis and Wager, 2019](#); [Efron, 2019](#)). Third, because latent group membership induces a finite-mixture distribution for individual effects, the paper also connects to finite-mixture models ([Dempster](#)

et al., 1977; McLachlan et al., 2019). Fourth, it contributes to research on econometric and statistical methods for network data (Zhu et al., 2017; Auerbach, 2022; Chen et al., 2021). The paper’s contribution is to integrate network-based group recovery with group-specific empirical Bayes shrinkage for two-way heterogeneity, and to characterize the resulting estimator’s identification and oracle-approximation properties.

The remainder of the paper is organized as follows. Section 2 presents the model and its conditional-dependence structure. Section 3 describes the group-learning and empirical Bayes procedures. Section 4 develops the identification and oracle-equivalence results. Section 5 reports the Monte Carlo study, and Section 6 presents the Amazon application. Section 7 concludes. The appendices contain proofs and supplementary material.

2 Two-Way Effect Regression Model with Group Structure

2.1 Main Model

In this subsection, we introduce the model. Let $\mathcal{G}_n = (\mathcal{S}_n, \mathcal{T}_n, \mathcal{E}_n)$ denote an observed bipartite network, where $\mathcal{S}_n$ and $\mathcal{T}_n$ are the two disjoint sets of nodes and $\mathcal{E}_n$ denotes the set of observed links. Let

$$N = |\mathcal{S}_n|, \quad M = |\mathcal{T}_n|, \quad n = |\mathcal{E}_n|.$$

For every observed link $(i, j) \in \mathcal{E}_n$, consider the two-way effect model for matched data

$$y_{ij} = \alpha_i^s + \alpha_j^t + x'_{ij}\beta_0 + \varepsilon_{ij},$$

where α_i^s denotes the heterogeneous effect associated with node $i \in \mathcal{S}_n$, α_j^t denotes the heterogeneous effect associated with node $j \in \mathcal{T}_n$, x_{ij} is a vector of observed covariates, and ε_{ij} is an idiosyncratic disturbance.

Stacking observations over all links gives

$$Y = B\alpha + X\beta_0 + \varepsilon, \tag{1}$$

where

$$\alpha = \begin{pmatrix} \alpha^s \\ \alpha^t \end{pmatrix} = \left(\alpha_1^s, \dots, \alpha_N^s, \alpha_1^t, \dots, \alpha_M^t \right)',$$

and B is the $n \times (N + M)$ bipartite incidence matrix. Each row of B corresponds to one observed link and contains a one in the column associated with its $\mathcal{S}_n$ -side node and a one in the column associated with its $\mathcal{T}_n$ -side node. Thus, if observation e corresponds to the link (i, j) ,

$$(B\alpha)_e = \alpha_i^s + \alpha_j^t.$$

Following the network two-way effect representation, define

$$\tilde{Y} = Y - X\beta_0.$$

Conditional on the true regression coefficient β_0 , the model becomes

$$\tilde{Y} = B\alpha + \varepsilon. \quad (2)$$

We assume that the incidence matrix B exhibits a matching pattern. Since B is the matrix representation of the underlying bipartite network, we use B and the network interchangeably throughout the paper. Figure 1 illustrates this structure. Panel (a) shows a bipartite network in which units interact more frequently with units from corresponding groups—for example, productive workers are more likely to be employed by efficient firms and vice versa. Panel (b) presents the corresponding adjacency matrix after reordering nodes by group membership, revealing a clear block structure associated with the matching pattern on both sides of the network.

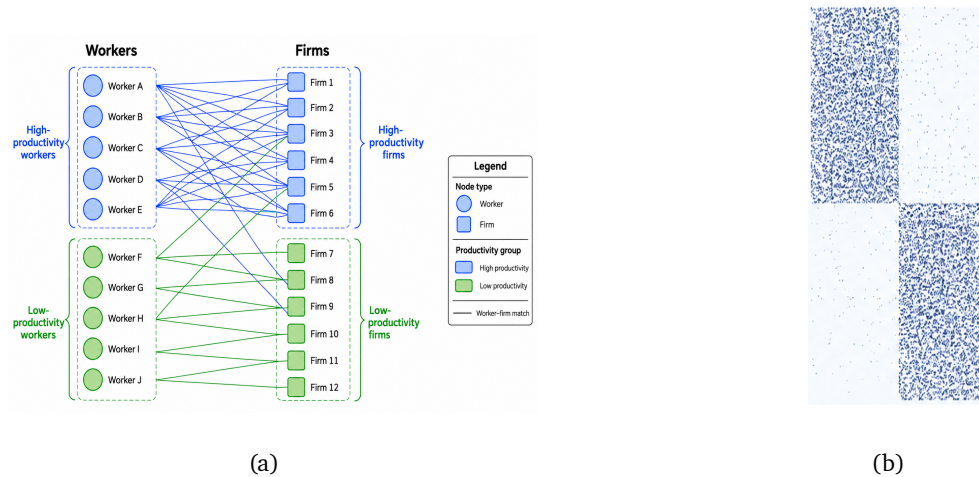


Figure 1: Panel (a): a bipartite network with matching pattern on both sides. Panel (b): the heatmap of the corresponding adjacency matrix after reordering nodes by group membership, showing the resulting block structure.

Furthermore, we assume that the two sets of heterogeneous effects possess potentially different latent group structures. In particular, suppose that the $\mathcal{S}_n$ side contains K latent groups and the $\mathcal{T}_n$ side contains L latent groups. Let

$$g_i \in \{1, \dots, K\}, \quad h_j \in \{1, \dots, L\}$$

denote the respective memberships.

Rather than imposing a common fixed effect on all units belonging to the same group, we assume that group membership determines the distribution of individual heterogeneity. Specifically,

$$\begin{aligned} \alpha_i^s \mid g_i = k &\sim \mathcal{N}(\mu_k^s, \tau_{k,s}^2), \quad k = 1, \dots, K, \\ \alpha_j^t \mid h_j = \ell &\sim \mathcal{N}(\mu_\ell^t, \tau_{\ell,t}^2), \quad \ell = 1, \dots, L. \end{aligned}$$

Hence units belonging to the same group share the same prior distribution but are not required to have identical heterogeneous effects. Equivalently, we may write

$$\begin{aligned} \alpha_i^s &= \mu_{g_i}^s + u_i^s, \quad \mathbb{E}[u_i^s \mid g_i] = 0, \\ \alpha_j^t &= \mu_{h_j}^t + u_j^t, \quad \mathbb{E}[u_j^t \mid h_j] = 0. \end{aligned}$$

Define the vector of group-specific prior means as

$$\mu_g = \left(\mu_1^s, \dots, \mu_K^s, \mu_1^t, \dots, \mu_L^t \right)',$$

and the vector of group-specific prior variances as

$$\tau_g^2 = \left((\tau_1^s)^2, \dots, (\tau_K^s)^2, (\tau_1^t)^2, \dots, (\tau_L^t)^2 \right)'.$$

Let the full set of prior and disturbance-variance parameters be

$$\psi = (\mu_g', (\tau_g^2)', \sigma_\varepsilon^2)'$$

Define the membership matrices

$$Z_s \in \{0, 1\}^{N \times K}, \quad Z_t \in \{0, 1\}^{M \times L},$$

where every row contains exactly one nonzero element indicating the group membership for unit i : for instance, if unit i belongs to group 2 out of three groups, the corresponding row of Z_s is $[0, 1, 0]$. The joint membership matrix is

$$Z = \begin{pmatrix} Z_s & 0 \\ 0 & Z_t \end{pmatrix}.$$

The vector of unit-specific prior means, $\mu = (\mu_{g_1}^s, \dots, \mu_{g_N}^s, \mu_{h_1}^t, \dots, \mu_{h_M}^t)'$, can be written compactly as $\mu = Z\mu_g$. Note that $N + M \gg K + L$, where K is the number of groups among the N units on the first side and L is the number of groups among the M units on the second side. An analogous representation applies to the unit-specific prior variances: $\tau^2 = Z\tau_g^2$, where $\tau^2 = (\tau_{g_1}^2, \dots, \tau_{g_N}^2, \tau_{h_1}^2, \dots, \tau_{h_M}^2)'$. Hence, conditional on the group memberships Z , the empirical Bayes estimation problem reduces from estimating the $(N + M)$ -dimensional vectors of unit-specific prior parameters μ and τ^2 to estimating the $(K + L)$ -dimensional vectors of group-specific means and variances, together with the common disturbance variance σ_ε^2 . In addition to the main specification, we propose a directed acyclic graph (DAG) that represents the underlying data-generating mechanism. Figure 2 illustrates the proposed generative structure and the associated dependence relationships among the model components. The DAG provides a graphical representation of the relationships among the observed outcomes $\tilde{Y}$, network links B , latent effects α , group memberships Z , the prior parameters (μ_g, τ_g^2) , and the disturbance variance σ_ε^2 .

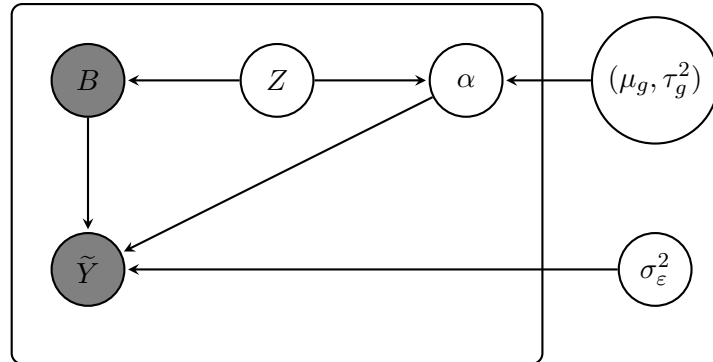


Figure 2: DAG assists with proposed model. Circle outside the plate denote hyperparameters, whereas circles inside the plates represent latent or observed variables. Shaded (gray) circles indicate observed quantities. In particular, the network B and the outcome $\tilde{Y}$ are observed.

A directed acyclic graph can be associated with a probabilistic graphical representation. The corresponding Bayesian network expresses the joint distribution through conditional distributions implied by the parent–child relationships. In particular, Figure 2 implies a factorization of the joint distribution of these variables into a product of lower-dimensional conditional distributions. This factorization is useful for estimation because the posterior distribution of the heterogeneous effects can be obtained from the components of the joint distribution that depend on those effects. Consequently, the DAG plays two roles in our framework: it provides a graphical representation of the dependence structure of the model and, at the same time, organizes the probabilistic factorization that leads to the proposed empirical-Bayes estimator.

Conditional on fixed group counts and hyperparameters, the maintained joint model is

$$\mathbb{P}(\tilde{Y}, B, \alpha, Z) = \mathbb{P}(Z) \mathbb{P}(\alpha | Z) \mathbb{P}(B | Z) \mathbb{P}(\tilde{Y} | B, \alpha). \quad (3)$$

Here the Gaussian prior has mean $Z\mu_g$ and diagonal covariance $D = \text{diag}(Z\tau_g^2)$, so effects are conditionally independent across nodes given Z and the hyperparameters. The factorization also states $B \perp\!\!\!\perp \alpha | Z$: matching may depend on group membership, but not on within-group effect innovations. We leave $\mathbb{P}(B | Z)$ otherwise unrestricted, subject to the group-recovery conditions below. The DAG is a representation of these assumptions, not an additional source of identification. The full posterior distribution of α , conditional on B and $\tilde{Y}$, is

$$\mathbb{P}(\alpha | \tilde{Y}, B) \propto \sum_Z \mathbb{P}(\tilde{Y}, B, \alpha, Z). \quad (4)$$

The full posterior in (4) averages over the unknown group memberships. The proposed feasible procedure uses a computationally simpler plug-in predictor: it estimates the memberships from the network, estimates the group-specific prior parameters, and then computes the posterior mean conditional on those estimates. This plug-in posterior is a working posterior and is not asserted to equal the full posterior that averages over Z . Its asymptotic relation to the oracle posterior mean is established in Section 4. The details of the estimation procedure are deferred to the next section. In the remainder of this section, we characterize the conditional distributions in Equation 3. When β_0 is known, the plug-in empirical-Bayes predictor is

$$\hat{\alpha}^{EB} = \mathbb{E}_{\hat{\psi}}[\alpha | \tilde{Y}, B, \hat{Z}]$$

and uses the estimated normalized hyperparameters $\hat{\psi}$. When β_0 is estimated, the feasible procedure replaces $\tilde{Y} = Y - X\beta_0$ with $\hat{Y} = Y - X\hat{\beta}$; Section 4.5 gives the corresponding oracle-equivalence result.

Under

$$\tilde{Y} = B\alpha + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2 I_n),$$

implies

$$\tilde{Y} \mid B, \alpha \sim \mathcal{N}(B\alpha, \sigma_\varepsilon^2 I_n).$$

Likewise, under the Gaussian group-specific prior,

$$\alpha \mid Z \sim \mathcal{N}(Z\mu_g, D(Z\tau_g^2)),$$

where $D(Z\tau_g^2)$ denotes the diagonal matrix of prior variances, and $Z\mu_g$ denotes the vector of prior means as defined above. We leave the conditional distribution of the network B given the group memberships Z largely unspecified. This allows flexibility in the network-formation mechanism while maintaining the assumption that Z generates the matching patterns used to recover group memberships. The network model must satisfy the regularity conditions for group recovery introduced below. A co-stochastic block model is one possible specification (Rohe et al., 2016).

The conditional independence assumption has substantive economic content: given group memberships, matching patterns provide no additional information about individuals' deviations from their group-specific effect means. For example, among workers in the same group, a worker's productivity deviation from the group mean is assumed not to predict which firms employ them. If such within-group sorting is important, the model would need to allow dependence between network formation and individual effect deviations.

3 Machine-Learning-Assisted Empirical Bayes Estimation

The feasible estimator first estimates the membership matrix Z and the normalized group-specific hyperparameters. It then computes the posterior mean conditional on these estimates. The resulting plug-in posterior is a working posterior; it is not the full posterior that averages over uncertainty in Z . Exact recovery of every label is not required for the oracle-equivalence result in Section 4.4: that result permits the fraction of misclassified nodes to vanish, provided

the additional likelihood-stability condition in Assumption 9 also holds. The estimator is

$$\hat{\alpha}^{EB} = \mathbb{E}_{\hat{\psi}} \left[\alpha \mid \hat{Y}, B, \hat{Z} \right], \quad (5)$$

where $\hat{\psi}$ denotes the estimated normalized prior parameters and disturbance variance.

3.1 Detecting Z Using Machine Learning

In this subsection, we discuss methods for estimating the latent membership matrix Z from the observed bipartite network B , and we denote the resulting estimate by $\hat{Z}$. Conditional on this estimated membership matrix, the working posterior satisfies

$$\mathbb{P}(\alpha \mid \hat{Y}, B, \hat{Z}) \propto \mathbb{P}(\hat{Y} \mid \alpha, B) \mathbb{P}(\alpha \mid \hat{Z}),$$

where $\mathbb{P}(\hat{Y} \mid \alpha, B)$ is the working likelihood based on the residualized outcome and $\mathbb{P}(\alpha \mid \hat{Z})$ is the group-specific prior given estimated group membership. The details of derive working posterior from equation (4) is left to Appendix SM.1.

Because group labels are identifiable only up to permutation, recovery is evaluated after optimal label alignment. The theory in Section 4.2 requires the misclassification proportions on both sides to converge to zero; it does not require exact recovery of every node. Hyperparameter consistency additionally requires the likelihood-stability condition in Assumption 9. Our framework can accommodate any network-clustering algorithm satisfying the embedding and clustering conditions stated there. One implementation uses spectral co-clustering: left and right singular vectors of a suitably normalized bi-adjacency matrix provide embeddings for the two node sets, which are clustered separately (Ng et al., 2001; Rohe et al., 2011, 2016).

3.2 The Estimation

In this subsection, we construct the empirical Bayes estimator.

The first step is to estimate the numbers of latent groups on the two sides of the network, denoted by K and L , respectively. These quantities determine the numbers of group-specific prior distributions: K distributions for the effects on one side of the network and L distributions for those on the other. Various procedures have been proposed for selecting the number of network communities (Le and Levina, 2015; Li et al., 2022; Karrer and Newman, 2011; Gao et al., 2018).

Because community-number selection is not the primary focus of this paper, our framework permits any computationally feasible and theoretically justified selection procedure.

The second step is to estimate the slope coefficients β from a preliminary two-way effect regression that controls for the node effects represented by B . In particular, letting

$$M_B = I - B(B'B)^\dagger B',$$

where $(B'B)^\dagger$ denotes the Moore–Penrose generalized inverse, we compute

$$\hat{\beta} = (X'M_B X)^{-1} X'M_B Y$$

and construct the residualized outcome

$$\hat{Y} = Y - X\hat{\beta}.$$

Here, $\hat{\beta}$ is a nuisance parameter estimated in the preliminary stage, while X is an observed covariate matrix.

The third step is to estimate the latent membership matrix Z from the network B , yielding $\hat{Z}$. Any network-clustering procedure is admissible provided that it satisfies the recovery condition required by the subsequent empirical Bayes analysis. Although our implementation uses spectral clustering, the empirical Bayes framework itself does not depend on a particular clustering algorithm.

Finally, given $\hat{Y}$, B , and $\hat{Z}$, we construct the empirical Bayes estimator. Under the Gaussian prior specification, we first estimate the group-level hyperparameters

$$\psi = (\mu_g, \tau_g^2, \sigma_\varepsilon^2)$$

by maximizing the marginal likelihood criterion. Denoting the resulting estimator by $\hat{\psi}$, we substitute $\hat{\psi}$ and $\hat{Z}$ into the posterior-mean expression to obtain the final empirical Bayes estimate of the heterogeneous node effects. Because $\hat{Y}$ uses an estimated regression coefficient, the criterion is a working marginal likelihood; Section 4.5 states conditions under which this substitution preserves oracle equivalence. We summarize the steps in Algorithm 1. Details of the estimation can be found in Appendix SM.2.

Algorithm 1 ML-Assisted Empirical Bayes Estimation

Input: outcome vector Y , covariate matrix X , and incidence matrix B .

- 1: Determine the number of groups $\widehat{K}$ and $\widehat{L}$ in B using a suitable group-selection procedure.
- 2: Using OLS, estimate $\widehat{\beta}$ and calculate $\widehat{Y} = Y - X\widehat{\beta}$.
- 3: Estimate the group membership structure $\widehat{Z}$ in B .
- 4: Estimate $\widehat{\psi}$ by maximizing the working marginal likelihood $\mathbb{P}(\widehat{Y} \mid B, \widehat{Z}, \psi) = \int_{\alpha} \mathbb{P}(\widehat{Y} \mid \alpha, B) \mathbb{P}(\alpha \mid \widehat{Z}, \psi) d\alpha$.
- 5: Compute the posterior mean using $\widehat{\psi}$: $\widehat{\alpha}^{EB} = \mathbb{E}(\alpha \mid B, \widehat{Y}, \widehat{Z}, \widehat{\psi})$.

Output: ML-Assisted Empirical Bayes estimator $\widehat{\alpha}^{EB} = \mathbb{E}(\alpha \mid B, \widehat{Y}, \widehat{Z}, \widehat{\psi})$.

4 Theory

This section develops identification conditions and asymptotic results for the proposed machine-learning-assisted empirical Bayes (ML-EB) estimator. The analysis proceeds in three steps. First, we derive conditions for identifying the two-way heterogeneous effects and the associated group-specific prior parameters in the bipartite-network setting. Second, we provide conditions under which the latent group memberships on both sides of the network can be consistently recovered. Third, we establish consistency of the estimated prior hyperparameters and show that, under consistent group recovery and hyperparameter estimation, the feasible empirical Bayes estimator asymptotically approaches its oracle counterpart.

4.1 Identification

This subsection distinguishes identification of the two-way effects through the outcome likelihood from identification of the group-specific prior parameters. The individual effects are treated as latent random variables, and their empirical Bayes estimation is a prediction problem under the specified hierarchical model. Accordingly, a uniquely defined posterior mean does not require the unrestricted realized effect vector to be point identified by the outcome likelihood alone.

Throughout this subsection, we condition on the observed covariates X , suppressing this conditioning where convenient. Identification of the prior parameters is considered conditional on the true membership matrix Z . Recovery of the unknown memberships is studied in the

following subsection.

Location indeterminacy of the outcome equation. Consider the model as defined in Section 2.1. Conditional on β_0 , define

$$Y_0 = Y - X\beta_0.$$

The residualized outcome equation is

$$Y_0 = B\alpha + \varepsilon.$$

Each row of B contains one unit entry corresponding to an $\mathcal{S}_n$ -side node and one unit entry corresponding to an $\mathcal{T}_n$ -side node. For an observed link (i, j) , the conditional mean therefore depends on the sum

$$\alpha_i^s + \alpha_j^t.$$

For any constant $d \in \mathbb{R}$, the transformation

$$\alpha_i^{s,*} = \alpha_i^s + d, \quad \alpha_j^{t,*} = \alpha_j^t - d \quad (6)$$

leaves this sum unchanged.

Define

$$v_0 = \begin{pmatrix} \mathbf{1}_N \\ -\mathbf{1}_M \end{pmatrix}. \quad (7)$$

Then $Bv_0 = 0$, so that

$$B(\alpha + dv_0) = B\alpha.$$

Consequently, the outcome likelihood cannot distinguish effect vectors that differ only by such a location shift.

Location normalization of the prior means. Under the maintained conditional Gaussian specification,

$$\alpha \mid B, Z \sim \mathcal{N}(m, D), \quad m = Z\mu_g, \quad D = \text{diag}(Z\tau_g^2).$$

The group means inherit a location indeterminacy. Adding a constant to all $\mathcal{S}_n$ -side group means and subtracting it from all $\mathcal{T}_n$ -side group means leaves the marginal mean $BZ\mu_g$ unchanged. We

impose the following restriction on the group-mean parameterization.

Assumption 1 (Location normalization of prior means). *The group means on the $\mathcal{T}_n$ side satisfy*

$$\sum_{\ell=1}^L \pi_{\ell}^t \mu_{\ell}^t = 0, \quad (8)$$

where

$$\pi_{\ell}^t = \frac{M_{\ell}}{M}, \quad M_{\ell} = \sum_{j=1}^M \mathbf{1}\{h_j = \ell\}.$$

Assumption 1 fixes the location of the prior means. Under the maintained conditional prior, it implies

$$\mathbb{E} \left[\frac{1}{M} \sum_{j=1}^M \alpha_j^t \middle| B, Z \right] = 0.$$

It does not require

$$\frac{1}{M} \sum_{j=1}^M \alpha_j^t = 0$$

for every realization.

We therefore retain the independent Gaussian specification for the underlying effects. No exact sample-centering restriction is imposed on the realized vector α . The normalization of prior means and identification of realized effects through the outcome likelihood are distinct issues.

Network connectivity and identified contrasts. Let C_n denote the number of connected components of the observed bipartite graph. For the incidence matrix used here,

$$\text{rank}(B) = N + M - C_n.$$

Each connected component therefore contributes one location indeterminacy.

Assumption 2 (Network connectivity). *The observed bipartite graph is connected. Consequently,*

$$\text{rank}(B) = N + M - 1, \quad (9)$$

and

$$\ker(B) = \text{span}\{v_0\}.$$

Proposition 1 (Identification up to a location shift). *Suppose Assumption 2 holds. Conditional on the regression coefficient, let $\alpha_1, \alpha_2 \in \mathbb{R}^{N+M}$ satisfy*

$$B\alpha_1 = B\alpha_2.$$

Then there exists a scalar $d \in \mathbb{R}$ such that

$$\alpha_1 - \alpha_2 = dv_0.$$

Consequently, the conditional mean of the outcome equation identifies within-side contrasts and cross-side sums, but does not separately identify the absolute locations of the two sets of realized effects.

More generally, a linear functional $a'\alpha$ is invariant to the location shift whenever $a'v_0 = 0$. Examples include

$$\alpha_i^s - \alpha_{i'}^s, \quad \alpha_j^t - \alpha_{j'}^t, \quad \alpha_i^s + \alpha_j^t.$$

Identification of these quantities does not imply that they can be estimated precisely. Their estimation accuracy depends on the amount and configuration of information supplied by the observed network. These are the outcome-identified functionals used to describe heterogeneity. The uncentered level of either side is not separately identified. We nevertheless predict the full effect vector under the normalized hierarchical prior; its posterior mean uses both the outcome likelihood and the prior to allocate location along the unidentified direction. This prediction target is distinct from likelihood identification of the realized effects.

Identification of regression coefficients. To distinguish β_0 from unrestricted node effects, the covariates must contain variation outside the column space of B . Define

$$P_B = B(B'B)^\dagger B', \quad M_B = I_n - P_B,$$

where † denotes the Moore–Penrose inverse.

Assumption 3 (Covariate identification). *Let $q = \dim(\beta_0)$. The residualized covariate matrix satisfies*

$$\text{rank}(M_B X) = q.$$

For the subsequent asymptotic analysis, suppose additionally that there exists a constant $c_X > 0$ such that

$$\lambda_{\min} \left(\frac{1}{n} X' M_B X \right) \geq c_X$$

with probability approaching one.

The rank condition is sufficient to identify β_0 separately from the unrestricted node effects. Indeed, if

$$B\alpha_1 + X\beta_1 = B\alpha_2 + X\beta_2,$$

premultiplication by M_B gives

$$M_B X(\beta_1 - \beta_2) = 0.$$

The rank condition implies $\beta_1 = \beta_2$. Under connectedness, the remaining difference between the effect vectors is then a location shift:

$$\alpha_1 - \alpha_2 = dv_0.$$

The lower eigenvalue bound strengthens the rank condition for asymptotic analysis. Establishing a convergence rate for an estimator of β_0 additionally requires appropriate assumptions on the regression disturbances.

Remark 1 (Common intercept). *A constant vector lies in the column space of B . A separate unrestricted intercept therefore cannot satisfy Assumption 3 when the node effects are otherwise unrestricted.*

The baseline specification absorbs the intercept into the node effects. If a separate intercept is introduced in the hierarchical model, the marginal-mean parameterization requires an additional location restriction. One possible convention is to center the prior means on both sides:

$$\sum_{k=1}^K \frac{N_k}{N} \mu_k^s = 0, \quad \sum_{\ell=1}^L \frac{M_\ell}{M} \mu_\ell^t = 0.$$

These are restrictions on the prior means, not exact restrictions on every realization of the individual effects. Identification and estimation of such an intercept must be treated separately from the residualized-covariate argument above.

Identification of group-specific prior means. Conditional on the true memberships, the marginal mean of the residualized outcome is

$$\mathbb{E}[Y_0 \mid B, Z] = BZ\mu_g.$$

Let

$$G = BZ.$$

Each row of G records the memberships of the two nodes associated with an observed link.

Define the observed group-level bipartite graph by connecting group k on the $\mathcal{S}_n$ side to group ℓ on the $\mathcal{T}_n$ side whenever at least one observed link joins nodes in these groups.

Assumption 4 (Group-level connectedness). *Every group contains at least one observed node, and the observed group-level bipartite graph is connected.*

Under Assumption 4,

$$\text{rank}(G) = K + L - 1,$$

and

$$\ker(G) = \text{span} \left\{ \begin{pmatrix} \mathbf{1}_K \\ -\mathbf{1}_L \end{pmatrix} \right\}.$$

Thus, two group-mean vectors generating the same marginal mean can differ only by adding a common constant to the means on one side and subtracting it from those on the other.

Assumption 1 removes this remaining indeterminacy. To see this, suppose that $\mu_{g,1}$ and $\mu_{g,2}$ generate the same marginal mean and both satisfy the normalization. Their difference must have the form

$$\mu_{g,1} - \mu_{g,2} = d \begin{pmatrix} \mathbf{1}_K \\ -\mathbf{1}_L \end{pmatrix}.$$

Because the product-side group shares sum to one, the two normalization restrictions imply $d = 0$. The normalized group means are therefore identified from the marginal mean.

When the individual-level graph is connected and all groups are nonempty, the observed group-level graph is also connected. In that case, Assumption 4 follows from Assumption 2.

Group-level connectedness is an identification condition. Consistent estimation additionally requires sufficient growth in the information distinguishing the group means.

Identification of variance components. Suppose that, conditional on B, Z ,

$$\varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2 I_n)$$

and that ε is independent of α . For $r = 1, \dots, K + L$, let D_r be the diagonal matrix selecting nodes in prior group r . Then

$$D = \sum_{r=1}^{K+L} \tau_r^2 D_r.$$

The marginal covariance matrix is

$$\Omega = BDB' + \sigma_\varepsilon^2 I_n = \sum_{r=1}^{K+L} \tau_r^2 BD_r B' + \sigma_\varepsilon^2 I_n. \quad (10)$$

Assumption 5 (Variance-component identification). *Conditional on the observed design and true memberships, the matrices*

$$BD_1 B', \dots, BD_{K+L} B', I_n$$

are linearly independent. Equivalently,

$$a_0 I_n + \sum_{r=1}^{K+L} a_r BD_r B' = 0$$

implies

$$a_0 = a_1 = \dots = a_{K+L} = 0.$$

Assumption 5 ensures that distinct variance-component vectors generate distinct marginal covariance matrices. Together with identification of the normalized group means, it identifies the prior parameters and disturbance variance in the conditional Gaussian marginal model

$$Y_0 \mid B, Z \sim \mathcal{N}(BZ\mu_g, BDB' + \sigma_\varepsilon^2 I_n).$$

Repeated observations sharing a node provide a useful interpretation of variance-component identification. For two distinct observed links sharing an $\mathcal{S}_n$ -side node,

$$\text{Cov}(y_{0,ij}, y_{0,ij'} \mid B, Z) = \tau_{g_i, s}^2, \quad j \neq j'.$$

Similarly, for two distinct observed links sharing an $\mathcal{T}_n$ -side node,

$$\text{Cov}(y_{0,ij}, y_{0,i'j} \mid B, Z) = \tau_{h_j, t}^2, \quad i \neq i'.$$

Such covariance information helps distinguish heterogeneity variances from disturbance variance. Connectedness alone does not guarantee identification of all variance components.

The linear-independence condition concerns identification for a given design. Consistent estimation along a sequence of growing networks additionally requires sufficient information growth and the regularity conditions used in the subsequent likelihood analysis.

Posterior prediction of individual effects. The location indeterminacy of the outcome likelihood does not prevent posterior prediction under a proper prior. For fixed memberships and hyperparameters, suppose D is positive definite and $\sigma_\varepsilon^2 > 0$. The posterior precision matrix is

$$D^{-1} + \frac{B'B}{\sigma_\varepsilon^2}.$$

This matrix is positive definite because D^{-1} is positive definite and $B'B$ is positive semidefinite. Hence the posterior distribution is proper, with covariance

$$V = \left(D^{-1} + \frac{B'B}{\sigma_\varepsilon^2} \right)^{-1}$$

and uniquely defined mean

$$\mathbb{E}[\alpha \mid Y_0, B, Z] = m + DB' \left(BDB' + \sigma_\varepsilon^2 I_n \right)^{-1} (Y_0 - Bm).$$

The prior supplies distributional information about directions that the outcome likelihood does not identify. This does not make the unrestricted realized effects point identified by that likelihood.

The empirical Bayes procedure replaces the unknown memberships and hyperparameters by estimated values. Its target remains the underlying latent effect vector α , and the oracle benchmark is the posterior mean computed using the true memberships and hyperparameters. The subsequent oracle-equivalence analysis compares these two predictors under the same hierarchical model.

Remark 2 (Location convention for least squares). *Because B is rank deficient, an unrestricted least-squares coefficient vector requires a location convention to select a unique representative. Such a convention is a restriction used to define the least-squares estimator; it is not automatically an exact restriction on the latent effects generated by the prior.*

Any comparison of individual-effect mean squared errors must therefore state the least-squares convention explicitly. Least squares should not be described as conditionally unbiased for the unrestricted latent vector solely because a location convention has been imposed. Comparisons based on within-side contrasts or matched-pair sums avoid this location ambiguity.

4.2 Recovery of the latent group structure

We distinguish recovery of most memberships from recovery of every membership. The subsequent empirical Bayes analysis uses an average squared-error criterion and permits a vanishing fraction of nodes to be misclassified. Exact recovery is not required.

Let

$$Z_0 = \text{diag}(Z_{s0}, Z_{t0}), \quad \hat{Z} = \text{diag}(\hat{Z}_s, \hat{Z}_t), \quad p_n = N + M.$$

The baseline analysis treats the numbers of groups K and L as fixed and known. Let $\mathcal{P}_K$ and $\mathcal{P}_L$ denote the sets of permutation matrices of the corresponding dimensions. Define

$$\begin{aligned} r_{s,n} &= \min_{\Pi_s \in \mathcal{P}_K} \frac{\|\hat{Z}_s - Z_{s0}\Pi_s\|_F^2}{2N}, \\ r_{t,n} &= \min_{\Pi_t \in \mathcal{P}_L} \frac{\|\hat{Z}_t - Z_{t0}\Pi_t\|_F^2}{2M}, \\ r_n &= \frac{Nr_{s,n} + Mr_{t,n}}{p_n}. \end{aligned} \tag{11}$$

Each incorrectly classified row of a one-hot membership matrix contributes two to the squared Frobenius norm. Thus these quantities are misclassification proportions after optimal label alignment. Convergence of $r_{s,n}$ and $r_{t,n}$ to zero does not imply that the probability of classifying every node correctly tends to one.

The following conditions provide a modular sufficient condition for recovery using spectral embeddings followed by clustering. They must be verified for the particular network model, embedding, and clustering algorithm used in an implementation. In particular, they are not asserted to follow from distinct raw block-probability profiles alone.

Assumption 6 (Network embeddings and clustering). *Membership estimation uses only the network and, if needed, an algorithmic random seed U_n . Conditional on (B, Z_0) , the seed is independent of the latent effects and outcome disturbances. For the S_n side, suppose that:*

(i) *There is a population target embedding*

$$F_{s0} = Z_{s0}C_{s,n} \in \mathbb{R}^{N \times d_s}.$$

When $K \geq 2$, its group centers satisfy

$$\delta_{s,n} = \min_{k \neq k'} \|C_{s,n}(k, \cdot) - C_{s,n}(k', \cdot)\| > 0.$$

(ii) *The estimated embedding $\hat{F}_s$, constructed from the network, satisfies*

$$e_{s,n}^2 := \min_{O'O=I_{d_s}} \|\hat{F}_s - F_{s0}O\|_F^2, \quad \frac{e_{s,n}^2}{N\delta_{s,n}^2} \xrightarrow{p} 0.$$

(iii) *For a fixed $\kappa \geq 1$, the fitted membership and center matrices obey the approximation guarantee*

$$\|\hat{F}_s - \hat{Z}_s \hat{C}_s\|_F^2 \leq \kappa \inf_{Z_s, C_s} \|\hat{F}_s - Z_s C_s\|_F^2,$$

where the infimum is over one-hot membership matrices with K groups and compatible center matrices.

Analogous conditions hold for the $\mathcal{T}_n$ side, with $M, L, d_t, C_{t,n}, \delta_{t,n}, e_{t,n}$ in place of the corresponding S_n -side quantities. If a side has only one group, all nodes on that side are assigned to it, its classification error is zero, and its separation and embedding-accuracy conditions are omitted.

The approximation guarantee in this assumption is a condition on the clustering procedure, not an automatic property of a local solution returned by a clustering routine. Likewise, embedding accuracy must be measured relative to population-center separation. An unscaled bound on singular-vector error alone need not establish recovery.

Proposition 2 (Consistency of latent-group recovery). *Suppose Assumption 6 holds and the group-growth condition in Assumption 8 below holds. Then*

$$r_{s,n} \xrightarrow{p} 0, \quad r_{t,n} \xrightarrow{p} 0, \quad r_n \xrightarrow{p} 0.$$

For each side with at least two groups, the corresponding misclassification proportion satisfies, with probability tending to one,

$$r_{s,n} \leq 8(\kappa + 1) \frac{e_{s,n}^2}{N\delta_{s,n}^2}, \quad r_{t,n} \leq 8(\kappa + 1) \frac{e_{t,n}^2}{M\delta_{t,n}^2}.$$

Hereafter, memberships and group-indexed parameters are aligned using the corresponding label permutations. This is an analytical convention; the estimator does not require knowledge of the true labels. Parameter-space and optimization conditions below are stated in these aligned coordinates. Estimated partitions are taken to have nonempty groups, and ties in the analytical label alignment are resolved deterministically.

Remark 3 (Estimated numbers of groups). *If the procedure estimates K and L , the subsequent conclusions extend provided that*

$$\Pr(\widehat{K} = K, \widehat{L} = L) \rightarrow 1$$

and the stated conditions hold on that event. Consistency of a particular group-number selection rule requires separate justification.

4.3 Estimation and consistency of the prior hyperparameters

We maintain the conditional Gaussian model and identification conventions described in Section 4.1. Conditional on the observed design and true memberships,

$$Y_0 = Y - X\beta_0 = B\alpha + \varepsilon, \quad \alpha \sim \mathcal{N}(m_0, D_0), \quad \varepsilon \sim \mathcal{N}(0, \sigma_{\varepsilon,0}^2 I_n),$$

where the two Gaussian vectors are conditionally independent,

$$m_0 = Z_0\mu_{g0}, \quad D_0 = \text{diag}(Z_0\tau_{g0}^2).$$

Conditioning on X is suppressed throughout. This subsection and the next take β_0 as known; preliminary regression estimation is treated separately in Section 4.5.

A common parameterization of the location restriction. The product-side prior means satisfy the normalization from Section 4.1. Because group shares change when memberships are

estimated, we implement this restriction using a common vector of free parameters. For a candidate membership matrix Z , let

$$w_\ell(Z) = \frac{1}{M} \sum_{j=1}^M \mathbf{1}\{h_j(Z) = \ell\}.$$

Write $\nu^t = (\mu_1^t, \dots, \mu_{L-1}^t)'$ and set

$$\mu_L^t(\nu^t, Z) = -\frac{\sum_{\ell=1}^{L-1} w_\ell(Z) \nu_\ell^t}{w_L(Z)}. \quad (12)$$

For $L = 1$, this definition is understood as $\mu_1^t = 0$. The free parameter vector is

$$\xi = ((\mu^s)')', (\nu^t)', (\tau_g^2)', (\sigma_\varepsilon^2)',$$

and $\psi(\xi, Z)$ denotes the associated full vector $(\mu_g', (\tau_g^2)', \sigma_\varepsilon^2)'$. This parameterization enforces the same form of location normalization for both true and estimated memberships. The true free parameter is denoted by ξ_0 , and $\psi_0 = \psi(\xi_0, Z_0)$. Dependence of the full normalized means on sample group shares is suppressed in the notation.

For a candidate (ξ, Z) , define

$$m(\xi, Z) = Z\mu_g(\xi, Z), \quad D(\xi, Z) = \text{diag}(Z\tau_g^2),$$

$$\Omega(\xi, Z) = BD(\xi, Z)B' + \sigma_\varepsilon^2 I_n.$$

The marginal log likelihood, up to a parameter-independent constant, is

$$\ell_n(\xi; Z) = -\frac{1}{2} \log |\Omega(\xi, Z)| - \frac{1}{2} \{Y_0 - Bm(\xi, Z)\}' \Omega(\xi, Z)^{-1} \{Y_0 - Bm(\xi, Z)\}. \quad (13)$$

Let $\hat{\xi}$ be a measurable maximizer over Ξ , or an approximate maximizer satisfying

$$\sup_{\xi \in \Xi} \ell_n(\xi; \hat{Z}) - \ell_n(\hat{\xi}; \hat{Z}) = o_p(a_n), \quad (14)$$

$$\hat{\psi} = \psi(\hat{\xi}, \hat{Z}).$$

A numerical stationary point alone does not establish this optimization condition.

Assumption 7 (Parameter space). *The free parameter space Ξ is compact, the true parameter ξ_0*

is in its interior, and the free means are uniformly bounded. There exist constants independent of n such that every candidate obeys

$$0 < \underline{\tau}^2 \leq \tau_r^2 \leq \bar{\tau}^2 < \infty, \quad r = 1, \dots, K + L,$$

$$0 < \underline{\sigma}^2 \leq \sigma_\varepsilon^2 \leq \bar{\sigma}^2 < \infty.$$

The same bounds hold for the true parameters.

Assumption 8 (Group growth). As $N, M \rightarrow \infty$, the numbers of groups K, L are fixed and

$$\frac{N_k}{N} \xrightarrow{p} \pi_k^s > 0, \quad \frac{M_\ell}{M} \xrightarrow{p} \pi_\ell^t > 0$$

for every k and ℓ .

Under Proposition 2, estimated group shares converge to their true counterparts after alignment. In particular, $w_L(\widehat{Z})$ and $w_L(Z_0)$ are bounded away from zero with probability tending to one. Together with compactness, this ensures uniform boundedness of the full candidate prior means in (12) on that event.

For likelihood consistency, we use criterion contrasts. This avoids requiring the entire uncentered likelihood, whose parameter components may have different information scales, to converge to a finite limit under one normalization. Define

$$\Delta_n(\xi; Z) = \ell_n(\xi; Z) - \ell_n(\xi_0; Z).$$

Let $\mathcal{K}_n(\xi)$ be the conditional Kullback–Leibler divergence from the true marginal Gaussian model to the candidate model with true memberships. With $b_\xi = Bm(\xi, Z_0)$, $b_0 = Bm_0$, $\Omega_\xi = \Omega(\xi, Z_0)$, and $\Omega_0 = BD_0B' + \sigma_{\varepsilon,0}^2 I_n$, it equals

$$\begin{aligned} \mathcal{K}_n(\xi) = & \frac{1}{2} \left[\log \frac{|\Omega_\xi|}{|\Omega_0|} + \text{tr}(\Omega_\xi^{-1} \Omega_0) - n \right] \\ & + \frac{1}{2} (b_\xi - b_0)' \Omega_\xi^{-1} (b_\xi - b_0). \end{aligned} \tag{15}$$

Thus $\mathbb{E}[\Delta_n(\xi; Z_0) \mid B, Z_0] = -\mathcal{K}_n(\xi)$.

Assumption 9 (Likelihood separation, concentration, and classification stability). There is a deterministic sequence $a_n \rightarrow \infty$ such that:

(i) For every $\epsilon > 0$ for which the set $\{\xi \in \Xi : \|\xi - \xi_0\| \geq \epsilon\}$ is nonempty, there exists $c_\epsilon > 0$ such that

$$\Pr \left(\inf_{\xi \in \Xi : \|\xi - \xi_0\| \geq \epsilon} \mathcal{K}_n(\xi) \geq c_\epsilon a_n \right) \longrightarrow 1.$$

(ii) Oracle likelihood contrasts concentrate uniformly:

$$\sup_{\xi \in \Xi} \frac{|\Delta_n(\xi; Z_0) + \mathcal{K}_n(\xi)|}{a_n} \xrightarrow{p} 0. \quad (16)$$

(iii) Classification negligibly perturbs criterion contrasts:

$$\sup_{\xi \in \Xi} \frac{|\Delta_n(\xi; \widehat{Z}) - \Delta_n(\xi; Z_0)|}{a_n} \xrightarrow{p} 0. \quad (17)$$

Both criteria use the common free parameter ξ , with their respective membership-dependent mean normalization.

Condition (17) concerns the effect of classification error on the likelihood. It is an additional condition, not a consequence of a vanishing misclassification proportion established in this paper. Exact recovery is one sufficient case, but the condition also allows imperfect classification. Part (i) requires growing information in every identified hyperparameter direction; finite-sample identification alone is insufficient. The likelihood scale a_n must make all three conditions hold and cannot be chosen solely to satisfy a subsequent rate restriction. Group growth and connectedness alone do not establish Assumption 9.

Information scales and matrix-level verification. The scale a_n must reflect the weakest information rate among the free prior-mean, prior-variance, and disturbance-variance parameters. To make this requirement explicit, write

$$b_\xi = Bm(\xi, Z_0), \quad A_\xi = \Omega(\xi, Z_0)^{-1}, \quad \delta_\xi = b_\xi - b_0,$$

and define

$$C_\xi = \Omega_0^{1/2} (A_\xi - A_0) \Omega_0^{1/2}, \quad v_\xi = \Omega_0^{1/2} A_\xi \delta_\xi, \quad A_0 = \Omega_0^{-1}.$$

Conditional on (B, Z_0) , let $z_n = \Omega_0^{-1/2}(Y_0 - b_0) \sim \mathcal{N}(0, I_n)$. The centered oracle likelihood contrast has the exact representation

$$\Delta_n(\xi; Z_0) + \mathcal{K}_n(\xi) = -\frac{1}{2}\{z_n' C_\xi z_n - \text{tr}(C_\xi)\} + v_\xi' z_n. \quad (18)$$

In particular,

$$\text{Var}(\Delta_n(\xi; Z_0) \mid B, Z_0) = \frac{1}{2} \text{tr}(C_\xi^2) + \|v_\xi\|^2. \quad (19)$$

These expressions give design-conditional quantities for checking the likelihood scale and concentration condition.

The local information matrix provides a complementary diagnostic. Let $G_n = \partial b_\xi / \partial \xi' |_{\xi=\xi_0}$ and $\Omega_{n,r} = \partial \Omega(\xi, Z_0) / \partial \xi_r |_{\xi=\xi_0}$. The conditional Fisher information for ξ has entries

$$[\mathcal{J}_n]_{rs} = [G_n' \Omega_0^{-1} G_n]_{rs} + \frac{1}{2} \text{tr}(\Omega_0^{-1} \Omega_{n,r} \Omega_0^{-1} \Omega_{n,s}). \quad (20)$$

A common normalization a_n must be small enough for the KL separation lower bound and large enough to dominate the stochastic fluctuations in (21). The eigenvalue rate of $\mathcal{J}_n$ is a local diagnostic for the weakest identified direction; the trace bounds in Proposition 3 also check that no likelihood component operates on a much larger scale. If these rates are incompatible, the common-scale assumption is too strong and the parameter blocks require separate normalizations. Even a well-conditioned local information matrix does not exclude distant observationally equivalent values, so the global KL separation in part (i) of Assumption 9 remains necessary.

The following proposition records matrix-level sufficient conditions for the concentration and classification-stability parts of Assumption 9. They are conditional design conditions, stated in terms of B , the memberships, and the induced Gaussian covariance matrices. The model leaves the distribution of $B \mid Z_0$ unrestricted, so deriving these conditions from a particular degree or block-probability specification requires an additional network sampling model.

Proposition 3 (Sufficient matrix conditions for likelihood regularity). *Suppose Ξ is a compact subset of a fixed-dimensional Euclidean space, Assumptions 6, 7, and 8 hold, and $a_n \rightarrow \infty$. Suppose also that the KL separation condition in part (i) of Assumption 9 holds. For the oracle contrasts, assume*

$$\sup_{\xi \in \Xi} \left\{ \text{tr}(C_\xi^2) + \|v_\xi\|^2 \right\} = O_p(a_n), \quad \sup_{\xi \in \Xi} \|C_\xi\|_{\text{op}} = O_p(1), \quad (21)$$

and that the centered process in (18) is continuously differentiable with

$$\sup_{\xi \in \Xi} \|\nabla_{\xi} \{\Delta_n(\xi; Z_0) + \mathcal{K}_n(\xi)\}\| = O_p(a_n).$$

For the estimated-membership criterion, define

$$\begin{aligned} \widehat{b}_{\xi} &= Bm(\xi, \widehat{Z}), \quad \widehat{\Omega}_{\xi} = \Omega(\xi, \widehat{Z}), \quad Q_{\xi} = \Omega_{\xi}^{-1/2}(\widehat{\Omega}_{\xi} - \Omega_{\xi})\Omega_{\xi}^{-1/2}, \\ u_{\xi} &= \Omega_{\xi}^{-1/2}(\widehat{b}_{\xi} - b_{\xi}), \quad s_{\xi} = \Omega_{\xi}^{-1/2}(b_0 - b_{\xi}). \end{aligned}$$

Assume uniformly over $\xi \in \Xi$ that

$$cI_n \preceq I_n + Q_{\xi} \preceq CI_n, \tag{22}$$

$$\begin{aligned} \sup_{\xi \in \Xi} \|Q_{\xi}\|_* &= o_p(a_n), & \sup_{\xi \in \Xi} \|Q_{\xi}\|_{\text{op}} &= o_p(1), \\ \sup_{\xi \in \Xi} \|u_{\xi}\|^2 &= o_p(a_n), & \sup_{\xi \in \Xi} \|s_{\xi}\|^2 &= O_p(a_n), \end{aligned} \tag{23}$$

where $\|\cdot\|_*$ denotes the nuclear norm. Finally, assume the criterion perturbation $\ell_n(\xi; \widehat{Z}) - \ell_n(\xi; Z_0)$ is continuously differentiable and its gradient has supremum $O_p(a_n)$ over Ξ . Then parts (ii) and (iii) of Assumption 9 hold, so the full likelihood assumption follows.

The nuclear-norm condition in (23) weights a membership error by the change it induces in the marginal covariance, while u_{ξ} measures its effect on the marginal mean in the covariance metric. Thus a small unweighted fraction of incorrect labels is not, by itself, sufficient; the errors must also have a small aggregate effect on the likelihood. The proposition supplies checkable sufficient conditions once a network design is specified, but does not derive them from group growth or connectedness alone.

Proposition 4 (Consistency of the estimated group priors). *Suppose Assumptions 7–9 hold, the estimator satisfies (14), and $r_{s,n}, r_{t,n} \rightarrow_p 0$. Then, after aligning labels,*

$$\widehat{\xi} \xrightarrow{p} \xi_0, \quad \|\widehat{\psi} - \psi_0\| \xrightarrow{p} 0.$$

This proposition establishes consistency of the hyperparameters under the stated likelihood conditions. It does not require exact membership recovery. Verification of the uniform likelihood conditions under primitive network assumptions is a separate requirement for applying the result

to a specific asymptotic network model.

4.4 Oracle and feasible empirical Bayes estimators

The oracle knows the true memberships and hyperparameters and predicts the underlying random effect vector α . Its posterior mean is

$$\alpha^{EB,O} = m_0 + D_0 B' \Omega_0^{-1} (Y_0 - B m_0), \quad \Omega_0 = B D_0 B' + \sigma_{\varepsilon,0}^2 I_n. \quad (24)$$

The feasible estimator uses

$$\hat{m} = \hat{Z} \hat{\mu}_g, \quad \hat{D} = \text{diag}(\hat{Z} \hat{\tau}_g^2), \quad \hat{\Omega} = B \hat{D} B' + \hat{\sigma}_\varepsilon^2 I_n$$

and is defined by

$$\hat{\alpha}^{EB} = \hat{m} + \hat{D} B' \hat{\Omega}^{-1} (Y_0 - B \hat{m}). \quad (25)$$

These are predictors under the hierarchical model; no exact realization-specific centering restriction is imposed on α .

Stability under imperfect classification. The key argument controls average posterior-mean error without requiring $\|\hat{D} - D_0\|_{\text{op}} \rightarrow_p 0$. Define

$$\Lambda_0 = \sigma_{\varepsilon,0}^2 D_0^{-1}, \quad \hat{\Lambda} = \hat{\sigma}_\varepsilon^2 \hat{D}^{-1}, \quad H_0 = B' B + \Lambda_0, \quad \hat{H} = B' B + \hat{\Lambda}.$$

The normal equations for the two posterior means yield the identity

$$\hat{\alpha}^{EB} - \alpha^{EB,O} = \hat{H}^{-1} \left\{ \hat{\Lambda}(\hat{m} - m_0) + (\Lambda_0 - \hat{\Lambda})(\alpha^{EB,O} - m_0) \right\}. \quad (26)$$

Lemma 1 (Average posterior-mean perturbation). *Maintain the correctly specified conditional Gaussian model above. Suppose the true and estimated full hyperparameter vectors have bounded means and satisfy the positive variance bounds in Assumption 7, with probability tending to one. Membership estimates use only (B, U_n) , with the conditional independence of U_n specified in Assumption 6. Then, after aligning group labels,*

$$\frac{1}{p_n} \|\hat{\alpha}^{EB} - \alpha^{EB,O}\|^2 = O_p \left(r_n + \|\hat{\psi} - \psi_0\|^2 \right). \quad (27)$$

Neither exact membership recovery nor independence between $\hat{\psi}$ and Y_0 is required.

The lemma separates the effects of estimating memberships and hyperparameters. Misclassified nodes can have nonvanishing errors in their assigned prior variances. Their contribution is controlled through their proportion and conditional second moments, rather than a maximum error across nodes.

Theorem 1 (Oracle equivalence under imperfect group recovery). *Maintain the conditional Gaussian model and identifying conventions of Section 4.1. Suppose Assumptions 6–9 hold and the hyperparameter estimator satisfies (14). Then*

$$\frac{1}{p_n} \|\hat{\alpha}^{EB} - \alpha^{EB,O}\|^2 \xrightarrow{p} 0. \quad (28)$$

More specifically,

$$\frac{1}{p_n} \|\hat{\alpha}^{EB} - \alpha^{EB,O}\|^2 = O_p \left(r_n + \|\hat{\xi} - \xi_0\|^2 + r_{t,n}^2 \right). \quad (29)$$

The last term accounts for the use of estimated group shares in the product-side prior-mean normalization.

The theorem allows a growing number of incorrectly classified nodes, provided that their proportions vanish on both sides and the likelihood conditions support consistent hyperparameter estimation. It establishes negligible average distance from the correctly specified oracle. It does not establish consistency of every individual effect, finite-sample dominance over least squares, or robustness to prior misspecification. No additional restriction on N/M is needed merely to transfer this posterior-mean result, because the estimand remains the underlying effect vector.

4.5 Estimated regression coefficients

Theorem 1 takes the regression coefficient β_0 as known. We now allow it to be estimated before fitting the group-specific priors and constructing the empirical Bayes predictor. We maintain the conditional Gaussian model of Section 4.3, including conditioning on X . Memberships $\hat{Z}$ are constructed from the network and the permitted algorithmic random seed as in Assumption 6; they are not re-estimated using the residualized outcomes.

Use the common free parameter vector ξ , the full parameter map $\psi(\xi, Z)$, and the membership-dependent mean normalization defined in Section 4.3. To make the dependence on the outcome

explicit, write

$$\begin{aligned}\ell_n(\xi; Z, y) = & -\frac{1}{2} \log |\Omega(\xi, Z)| \\ & -\frac{1}{2} \{y - Bm(\xi, Z)\}' \Omega(\xi, Z)^{-1} \{y - Bm(\xi, Z)\}.\end{aligned}\tag{30}$$

Thus $\ell_n(\xi; Z, Y_0)$ is the criterion used in Section 4.3. Let $\tilde{\xi}$ be a measurable approximate maximizer of the criterion based on $\hat{Y}$, satisfying

$$\sup_{\xi \in \Xi} \ell_n(\xi; \hat{Z}, \hat{Y}) - \ell_n(\tilde{\xi}; \hat{Z}, \hat{Y}) = o_p(a_n), \quad \tilde{\psi} = \psi(\tilde{\xi}, \hat{Z}).\tag{31}$$

Exact maximization is a special case. Define the posterior-mean map

$$\begin{aligned}T_n(y; \xi, Z) = & m(\xi, Z) \\ & + D(\xi, Z) B' \Omega(\xi, Z)^{-1} \{y - Bm(\xi, Z)\}.\end{aligned}\tag{32}$$

The feasible predictor and the unchanged oracle benchmark are

$$\tilde{\alpha}^{EB} = T_n(\hat{Y}; \tilde{\xi}, \hat{Z}), \quad \alpha^{EB, O} = T_n(Y_0; \xi_0, Z_0).$$

Both predict the underlying random effect vector; no exact centering restriction is imposed on its realized values.

Assumption 10 (Preliminary regression estimation). *As $n, p_n \rightarrow \infty$, the dimension of β_0 is fixed and*

$$\frac{\|X\|_F^2}{n} = O_p(1), \quad \|\hat{\beta} - \beta_0\| = O_p(n^{-1/2}).$$

This assumption implies

$$\|e_n\| \leq \|X\|_F \|\hat{\beta} - \beta_0\| = O_p(1), \quad \frac{\|e_n\|^2}{n} = O_p(n^{-1}).$$

The aggregate residualization error need not vanish, but its average squared magnitude does. The root- n rate is a requirement on the preliminary estimator, not an automatic consequence of using least squares. Its justification requires suitable identification, exogeneity, and information-growth conditions for the regression.

Assumption 11 (Likelihood scale for residualization error). *For the deterministic sequence a_n in*

Assumption 9,

$$\frac{\sqrt{n}}{a_n} \rightarrow 0.$$

Under the maintained variance bounds and bounded prior means, estimating β_0 changes the marginal log likelihood uniformly by at most $O_p(\sqrt{n})$. This assumption makes that perturbation negligible relative to the separation scale used in Section 4.3. The sequence a_n must satisfy all parts of Assumption 9; it cannot be selected solely to satisfy the present restriction. If those conditions hold with $a_n \asymp p_n$, the additional restriction is $n = o(p_n^2)$.

Corollary 1 (Oracle equivalence with estimated regression coefficients). *Maintain the conditional Gaussian model and identification conventions of Theorem 1. Suppose Assumptions 6–9, 10, and 11 hold, and $\tilde{\xi}$ satisfies (31). Then, after label alignment,*

$$\tilde{\xi} \xrightarrow{p} \xi_0, \quad \|\tilde{\psi} - \psi_0\| \xrightarrow{p} 0,$$

and

$$\frac{\|\tilde{\alpha}^{EB} - \alpha^{EB,O}\|^2}{p_n} \xrightarrow{p} 0. \quad (33)$$

More specifically,

$$\frac{\|\tilde{\alpha}^{EB} - \alpha^{EB,O}\|^2}{p_n} = O_p\left(r_n + \|\tilde{\xi} - \xi_0\|^2 + r_{t,n}^2 + p_n^{-1}\right). \quad (34)$$

The result extends Theorem 1 to estimating both regression coefficients and prior hyperparameters. It permits a vanishing fraction of misclassified nodes, subject to the likelihood stability condition already imposed in Section 4.3. Exact membership recovery is not required. The term p_n^{-1} bounds the direct posterior-mean effect of residualization error. Estimating β_0 can also affect the hyperparameter error $\|\tilde{\xi} - \xi_0\|$; the displayed bound does not assert that this indirect effect is of order p_n^{-1} .

Remark 4 (Scope of the conclusion). *No sample splitting or independence between $\hat{\beta}$ and the outcomes is required. The Gaussian criterion evaluated at $\hat{Y}$ is generally a working criterion, rather than the exact finite-sample likelihood of $\hat{Y}$. The proof transfers likelihood separation from the correctly residualized outcome and then controls the posterior perturbation. The conclusion concerns average distance from the oracle; it does not by itself justify ignoring preliminary-estimation uncertainty in confidence intervals.*

Remark 5 (More general preliminary estimators). *The stated assumptions are convenient sufficient conditions. The same argument applies to other preliminary rates if*

$$\frac{\sqrt{n} \|e_n\| + \|e_n\|^2}{a_n} \xrightarrow{p} 0, \quad \frac{\|e_n\|^2}{p_n} \xrightarrow{p} 0.$$

The first condition controls the likelihood perturbation and the second controls the direct posterior perturbation. In the rate bound, p_n^{-1} is then replaced by $\|e_n\|^2/p_n$.

5 Monte Carlo Simulation

The proposed Monte Carlo study will evaluate whether learning latent groups from matching patterns improves estimation beyond ordinary empirical Bayes shrinkage. The design separates the availability of observations for individual nodes, the informativeness of the network about memberships, and the extent to which memberships predict the distribution of individual effects. Comparisons with infeasible benchmarks will distinguish the costs of estimating memberships, prior hyperparameters, and regression coefficients. Supplementary experiments will examine weak connectivity and departures from the maintained Gaussian group model.

5.1 Data-generating processes

Baseline model. The baseline has $N = M = 600$ nodes and $K = L = 3$ groups. On each side, exactly one-third of the nodes belong to each group, and node labels are randomly permuted. Memberships are generated before the network and the effects. Conditional on memberships, network formation is independent of the innovations in the individual effects. Thus, after integrating out auxiliary network propensities, the baseline obeys the factorization

$$p(Z, B, \alpha, Y \mid X) = p(Z)p(B \mid Z)p(\alpha \mid Z)p(Y \mid B, \alpha, X).$$

The effects are generated as

$$\begin{aligned} \alpha_i^s &= \mu_{g_i}^s + \tau_{g_i, s} u_i^s, & u_i^s &\stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1), \\ \alpha_j^t &= \mu_{h_j}^t + \tau_{h_j, t} u_j^t, & u_j^t &\stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1), \end{aligned} \tag{35}$$

with independence across both sides. Memberships are not redefined using the realized effects.

Table 1 specifies four configurations. Each listed mean vector and variance vector applies to both sides in the balanced baseline. The variance entries are $\tau_{k,s}^2$ and $\tau_{\ell,t}^2$, rather than standard deviations. The average within-group variance is 0.09 in all four configurations.

Table 1: Baseline configurations of the group-specific effect distributions

Configuration	Group means	Group variances
No distributional differences	(0, 0, 0)	(0.09, 0.09, 0.09)
Mean differences only	(−0.6, 0, 0.6)	(0.09, 0.09, 0.09)
Variance differences only	(0, 0, 0)	(0.0225, 0.09, 0.1575)
Mean and variance differences	(−0.6, 0, 0.6)	(0.0225, 0.09, 0.1575)

The first configuration is a negative control for the usefulness of group-specific shrinkage: network communities can be present even though all groups on a given side share the same effect distribution. In that case, estimating separate priors introduces additional parameters without additional distributional information.

Equal group shares imply that the specified product-side means have weighted average zero. For unequal group shares, preliminary values a_ℓ^t will be converted to

$$\mu_\ell^t = a_\ell^t - \sum_{q=1}^L w_q^t a_q^t, \quad w_q^t = M_q/M.$$

This restriction concerns prior means. The realized effects will not be recentered during data generation, since doing so would alter the independent Gaussian prior.

Network formation. We use a degree-corrected bipartite planted-partition model (Zhao et al., 2024). Let $A \in \{0, 1\}^{N \times M}$ be the bi-adjacency matrix. Before imposing connectedness, its entries are conditionally independent Bernoulli variables with probabilities

$$P_{ij} = \frac{d}{M} \theta_i \phi_j \{ (1 - \eta) + 3\eta \mathbf{1}\{g_i = h_j\} \}. \quad (36)$$

Within each group on each side, half the nodes are randomly assigned propensity 0.75 and half propensity 1.25. These assignments are independent of the effect innovations. The within-group mean propensities are one, so in the balanced design

$$\mathbb{E} \left[\sum_j A_{ij} \mid Z, \theta, \phi \right] = d\theta_i, \quad \mathbb{E} \left[\sum_i A_{ij} \mid Z, \theta, \phi \right] = d\phi_j.$$

Here d controls expected average degree, whereas η controls the strength of matching by group without changing these expected degrees. The main grid is

$$d \in \{12, 24, 48\}, \quad \eta \in \{0.25, 0.60, 0.90\}.$$

Together with Table 1, this gives 36 main cells. All probabilities in this grid lie strictly between zero and one. An additional $\eta = 0$ design will examine a network carrying no membership information beyond the independently generated propensities. The incidence matrix B is constructed from the realized edges, and $n = \sum_{i,j} A_{ij}$ is random.

For comparisons requiring connectedness, graphs will be sampled from (36) conditional on being connected, using rejection sampling. Acceptance rates and realized degree distributions will be reported. The displayed expected-degree formulas apply before this conditioning. Conditional sampling changes the graph distribution and does not preserve independent edges, but it preserves independence between the graph and effect innovations given memberships. Nodes will not be silently removed and additional edges will not be inserted to repair rejected draws. A computational cap and all unsuccessful sampling attempts will be documented.

Covariates and outcomes. For each observed edge, generate

$$x_{ij} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1), \quad \varepsilon_{ij} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1), \quad \beta_0 = 1,$$

independently of memberships, the graph, and effect innovations, and set

$$y_{ij} = \alpha_i^s + \alpha_j^t + x_{ij}\beta_0 + \varepsilon_{ij}. \quad (37)$$

There is no separate intercept in X . Two versions will be considered. The first uses the known β_0 , isolating membership and prior estimation. The second uses

$$\hat{\beta} = (X' M_B X)^{-1} X' M_B Y, \quad M_B = I - B(B' B)^{\dagger} B',$$

and residualizes the outcome by $Y - X\hat{\beta}$. All feasible procedures in the second version use the same preliminary estimate. The rank of $X' M_B X$ and regression-estimation failures will be recorded. The number of observations and residual degrees of freedom will also be reported.

Separating mean heterogeneity from total signal variation. As a supplementary experiment, set the mean vector on each side to $(-\delta, 0, \delta)$ and use a common within-group variance

$$\tau^2 = v - \frac{2}{3}\delta^2, \quad v = 0.36, \quad \delta \in \{0, 0.3, 0.6\}.$$

For a uniformly sampled node, total marginal effect variance is then v for every value of δ . This experiment separates the role of between-group mean differences from changes in overall effect variance.

5.2 Competing estimators and implementation

Table 2 summarizes the comparison. For the marginal-likelihood procedures, the free group means, prior variances, and disturbance variance will be estimated jointly, with the product-side prior means normalized using the corresponding estimated group shares.

Table 2: Estimators and the information available to each procedure

Estimator	Specification and purpose
Two-way OLS	Unrestricted node effects, with a declared location convention; benchmark for the overall value of regularization (Jochmans and Weidner, 2019).
Ordinary EB	One Gaussian prior per side, with distinct side-specific variances and estimated hyperparameters; benchmark without group learning.
Degree-group EB	Separate priors for groups formed from observed degree quantiles; benchmark for information in node activity.
Network-group EB	Estimated network memberships and estimated hyperparameters; the proposed feasible procedure.
True-group EB	True memberships with estimated hyperparameters; benchmark for the cost of learning memberships.
Known-prior EB	True memberships and hyperparameters, evaluated using $Y - X\hat{\beta}$ in the estimated-coefficient experiment.
Full oracle	True memberships, hyperparameters, and regression coefficient; the correctly specified oracle posterior mean.
CHS	The network-dependent shrinkage specification of Cheng et al. (2025), with hyperparameters selected by URE.

Ordinary EB uses $m = (\mu_s \mathbf{1}'_N, 0 \mathbf{1}'_M)'$ and $D = \text{diag}(\tau_s^2 I_N, \tau_t^2 I_M)$, where the zero product-side prior mean fixes location. It is fitted as a joint two-way random-effects model. It is not implemented by independently shrinking OLS coefficients while ignoring their estimation covariance. Degree-group EB uses K degree groups on the first side and L on the second, with ties broken using an outcome-independent random seed. It therefore has the same nominal number of group-specific prior parameters as network-group EB.

The main experiment treats K, L as known. Network-group estimation uses only A , never

the outcomes. For the balanced three-group design, a regularized spectral implementation will form $A_\rho = \text{diag}(d_i + \rho_s)^{-1/2} A \text{diag}(q_j + \rho_t)^{-1/2}$ where $\rho_s = n/N$ and $\rho_t = n/M$, retain the three leading left and right singular vectors, normalize nonzero embedding rows to unit length, and cluster each side separately. The leading singular vector is retained. Zero embedding rows, ties, and empty clusters require predetermined handling and will be recorded. The regularized co-clustering approach is motivated by [Rohe et al. \(2016\)](#). Multiple initializations will be used, retaining the smallest clustering objective. This numerical procedure does not by itself certify the approximation guarantee imposed in the theory.

The CHS comparison will implement that paper’s actual prior and URE criterion, including its treatment of unknown disturbance variance. Its location convention and the loss targeted by its tuning criterion will be documented. An arbitrary ridge estimator will not be labeled as CHS. Because CHS and the proposed estimator differ in both prior specification and hyperparameter selection, their comparison measures the performance of the complete procedures rather than isolating a single modeling component.

5.3 Main Result

The proposed estimator uses regularized co-spectral clustering of the bipartite adjacency matrix, followed by empirical Bayes estimation of the group-specific Gaussian prior parameters. In this study, the clustering step is supplied with the true group counts, $K = L = 3$. The comparators are OLS, ordinary (pooled) empirical Bayes, degree-group empirical Bayes, empirical Bayes with true group memberships, an estimator using the true prior moments, and the CHS URE procedure. We carry out 200 replications for each.

Figure 3 reports pair-sum mean squared error averaged equally over the 36 design cells, using the estimated- β_0 results. The averages are 0.1206 for OLS, 0.0762 for ordinary EB, 0.0766 for degree-group EB, and 0.0668 for co-spectral EB. Thus, co-spectral EB lowers pair-sum MSE by 44.6% relative to OLS, 12.3% relative to ordinary EB, and 12.7% relative to degree-group EB. The corresponding averages are 0.0607 for true-group EB, 0.0596 for known-moment EB, and 0.0799 for the CHS URE benchmark. Co-spectral EB has lower pair-sum MSE than ordinary EB in 24 of the 36 factorial cells. As a secondary criterion, centered within-side MSE averages 0.0603 for OLS, 0.0380 for ordinary EB, 0.0382 for degree-group EB, and 0.0334 for co-spectral EB; the true-group, known-moment, and CHS averages are 0.0303, 0.0298, and 0.0399, respectively.

The incremental value of network grouping depends on both prior heterogeneity and network

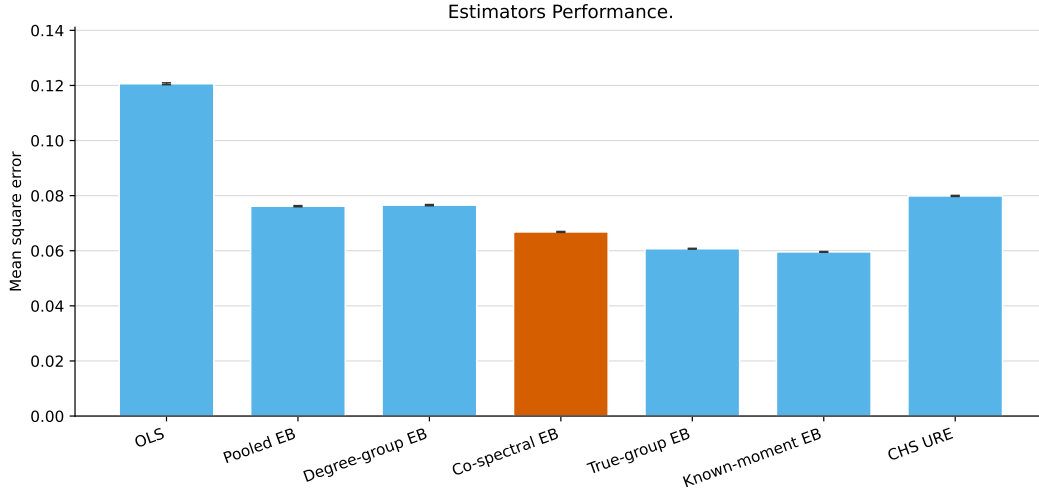


Figure 3: Pair-sum mean squared error across estimators. Bars are equally weighted averages over the 36 design cells for the estimated- β_0 case.

strength. Figure 4 plots the difference in pair-sum MSE between ordinary EB and co-spectral EB; positive values favor the proposed estimator. Averaged over degree, when group means and variances differ, the gain is 0.0018 at $\eta = 0.25$, 0.0248 at $\eta = 0.60$, and 0.0310 at $\eta = 0.90$. When group distributions are identical, the gain is small and negative at all three network-strength levels (between -0.0005 and -0.0016). This negative-control design shows that estimating separate group priors can add a small amount of error when the groups do not represent prior heterogeneity. Membership recovery follows the network signal: at $\eta = 0.25$, the average misclassification rate falls from 57.7% at degree 12 to 7.0% at degree 48; at $\eta = 0.60$, it is 3.0% at degree 12 and below 0.2% at degrees 24 and 48. The largest gains arise when groups are readily recovered and each node has relatively little outcome information, making group-specific shrinkage most useful.

The known- and estimated- β_0 results are nearly identical for pair-sum mean squared error in this design (0.066829 and 0.066833, respectively). This indicates little first-stage cost here, but does not establish that estimating β is negligible in sparser samples or under weaker regression identification. Additional checks vary the alignment between network groups and effect distributions, degree heterogeneity, prior misspecification, group shares, network size, and group-count selection. These results are reported in Appendix SM.4.

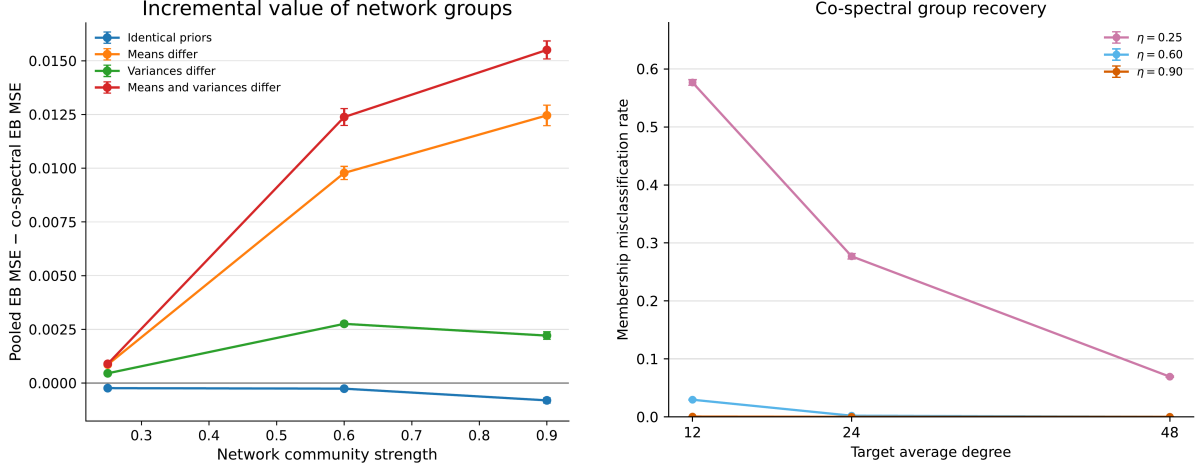


Figure 4: Incremental value and recovery of network groups. The left panel reports ordinary-EB pair-sum mean squared error minus co-spectral-EB pair-sum mean squared error, averaged over the three degree levels; positive values favor co-spectral EB. The right panel reports membership misclassification averaged across the two network sides and the four prior configurations.

6 Empirical Study

6.1 Amazon User Ratings

Online product ratings have become an important source of information for consumers, firms, and researchers studying digital markets. Existing research documents that Amazon ratings and reviews can affect product sales, consumers’ acquisition of information, and consumer welfare (Chevalier and Mayzlin, 2006; Reimers and Waldfogel, 2021). Other studies emphasize that observed ratings may also be shaped by reviewer identity, selective reviewing, and the sequential evolution of consumer opinions (Godes and Silva, 2012; Forman et al., 2008). Accordingly, an observed rating may reflect not only the underlying appeal of a product but also the systematic rating tendency of the user providing the evaluation: some users consistently assign relatively high ratings, whereas others evaluate products more critically. Nevertheless, much of the existing literature relies on product-level average ratings or focuses on particular characteristics of reviews, leaving the two-sided heterogeneity underlying individual user–product evaluations relatively underexplored. In this subsection, we represent Amazon ratings as observations generated on a bipartite network connecting users and products and estimate a model incorporating both user- and product-specific heterogeneity,

$$\text{Rating} = \alpha_0 + B\alpha + \varepsilon,$$

where B is the incidence matrix derived from Amazon User-Product rating bipartite network. Furthermore, α_0 denotes baseline and $\alpha = [\text{User}, \text{Product}]'$. The model decomposes each observed rating into a common component, a persistent user-specific rating tendency, a product-specific appeal component, and an idiosyncratic residual, thereby providing a more granular characterization of rating variation on both sides of the online marketplace.

Our empirical analysis uses the Industrial and Scientific category of the Amazon Reviews'23 dataset compiled by the McAuley Lab at the University of California, San Diego. Amazon Reviews'23 is a large-scale collection of product reviews and user-product interactions spanning numerous product categories and an extended period. The complete Industrial and Scientific category contains approximately 5.2 million ratings submitted by 3.4 million users for approximately 427,500 products. Rather than using these raw data, we employ the officially provided 5-core, de-duplicated (*Pure IDs*) version. The 5-core procedure iteratively removes users and products with fewer than five distinct interactions, thereby retaining a denser portion of the bipartite network in which every included node has repeated connections. De-duplication removes duplicate review records and retains a single observation for each included user-product pair. These preprocessing steps substantially mitigate the weak-connectivity problem and limit the influence of users and products that appear only infrequently. This restriction is particularly useful in our setting because estimation of user- and product-specific heterogeneity requires sufficient overlap between the two sides of the network, while the precision of node-specific estimates generally deteriorates for low-degree nodes. The resulting estimation sample contains 412,947 distinct user-product rating observations involving 50,985 unique users and 25,848 unique parent products. The reviews in the estimation sample were posted between July 2001 and August 2023.

We represent the data as an undirected and unweighted bipartite network. As in Section 2, let $B \in \{0, 1\}^{n \times (N+M)}$ denote the edge-by-node incidence matrix used to define the two-way effect model, where n is the number of observed links and N and M are the numbers of users and products, respectively. For the network-clustering step in this section, it is more convenient to work with the corresponding bi-adjacency matrix $A \in \{0, 1\}^{N \times M}$, and

$$A_{ij} = \mathbf{1}\{\text{user } i \text{ rated product } j\}.$$

Writing $B = [B_s \ B_t]$, where $B_s \in \{0, 1\}^{n \times N}$ and $B_t \in \{0, 1\}^{n \times M}$ collect the user- and product-

side columns of B , respectively, the two representations are related by $A = B'_s B_t$: since the data are de-duplicated, each observed user–product pair contributes exactly one link, so every entry of $B'_s B_t$ is either 0 or 1 and coincides with A_{ij} . In our Amazon Industrial and Scientific sample, the resulting network contains $N = 50,985$ users, $M = 25,848$ products, and $n = 412,947$ observed user–product links.

We select the numbers of user and product groups using the degree-corrected bipartite adaptation of the corrected BIC described in Appendix SM.5. For each candidate pair, the memberships are estimated from the network and the degree-corrected profile objective is evaluated. We then choose

$$(\widehat{K}, \widehat{L}) = \arg \min_{(K,L) \in \mathcal{C}} \text{CBIC}_{\text{DC}}(K, L),$$

where $\mathcal{C}$ is the prespecified set of candidate group-count pairs and the criterion is defined in Appendix SM.5. This is a bipartite adaptation of the CBIC principle; its penalty is not the one-mode formula of Hu et al. (2020) applied without modification.

Applying the group-number selection procedure to the empirical network yields $\widehat{K} = 10$ user groups and $\widehat{L} = 11$ product groups. Given these selected numbers of groups, we construct spectral embeddings from the left and right singular vectors of the degree-normalized bi-adjacency matrix and apply clustering separately to the user and product embeddings (Rohe et al., 2016). Figure 5 presents the resulting group structure. Panel (a) displays the user–product bipartite network. Panel (b) displays the bi-adjacency matrix after its rows and columns have been reordered according to the estimated user and product memberships, respectively. The reordered matrix exhibits pronounced blockwise variation in link density, indicating that particular user groups interact disproportionately with particular product groups. Each rectangular block represents the interactions between one user group and one product group.

Next, given estimated group memberships, we estimate unobserved heterogeneity. The empirical Bayes procedure exploits the estimated group structure of the user–product network by allowing the distributions of the user and product effects to vary across their respective latent groups. We adopt group-specific Gaussian priors for both sets of effects. Because users are partitioned into 10 groups and products into 11 groups, the specification contains 21 group-specific prior distributions: 10 for the user effects and 11 for the product effects. This structure accommodates heterogeneity both among individual users and products within the same group and across network-defined groups through differences in the locations and dispersions of their prior

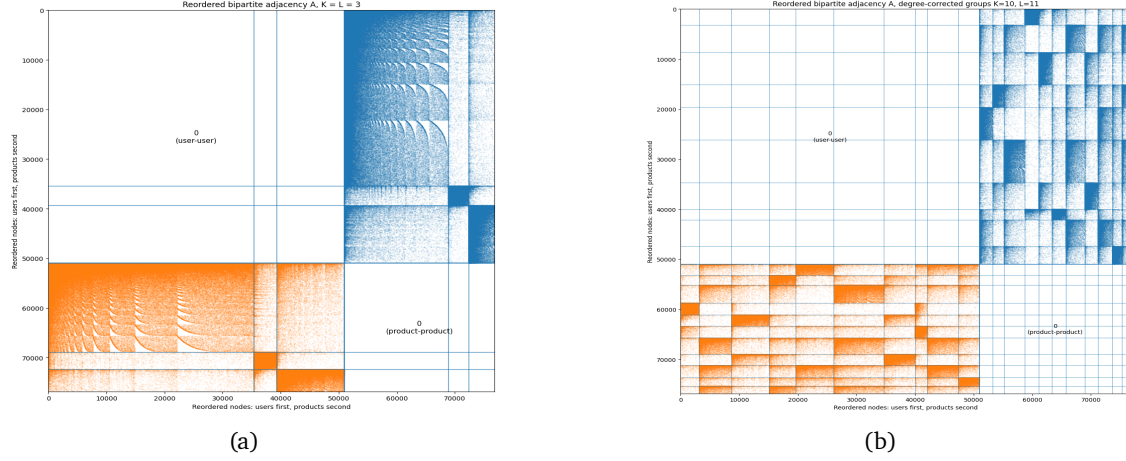


Figure 5: Panel (a) and (b) display the heatmaps of the corresponding bi-adjacency matrix, with both rows and columns reordered according to the estimated cluster memberships obtained from the bipartite spectral clustering procedure. In panel (a), both the numbers of groups in row and column are set to $K=L=3$. In panel (b), the number of groups in row is set to $K=10$ and the number of groups in column is set to $K=11$.

distributions. Under the normalization imposed on the user and product effects, the baseline α_0 is estimated using preliminary FE method. The estimated baseline $\hat{\alpha}_0$ is 4.425. The detail of estimation is left to Section SM.7. Table 3 reports the estimated group-specific prior means and standard deviations.

Table 3: Estimated group-specific prior parameters.

User Groups				Product Groups			
Group	N_{nodes}	Prior Mean	Prior SD	Group	N_{nodes}	Prior Mean	Prior SD
1	3,152	0.1444	0.1580	1	2,252	-0.0211	0.2651
2	5,531	0.0607	0.4068	2	1,903	0.0070	0.2434
3	6,429	-0.1709	0.6919	3	3,576	0.0642	0.2187
4	4,515	0.0353	0.4529	4	2,387	0.1207	0.0945
5	6,525	0.0152	0.4921	5	2,241	-0.1756	0.3819
6	8,558	0.0621	0.4136	6	2,375	-0.0119	0.2241
7	5,295	-0.0746	0.5927	7	3,270	0.0737	0.1997
8	2,085	0.0998	0.2980	8	2,188	-0.1789	0.3864
9	5,314	0.0321	0.4515	9	2,473	0.0081	0.2520
10	3,581	0.0120	0.4753	10	1,655	0.0163	0.2615
				11	1,528	0.0273	0.2334

Notes: Group means and variances are estimated jointly by marginal maximum likelihood, conditional on the supplied group memberships. The FE intercept and innovation variance are held fixed at $\hat{\beta}_0 = 4.425580$ and $\hat{\sigma}_\epsilon^2 = 0.846414$. Prior SD denotes the square root of the estimated prior variance. Product-group means satisfy the normalization $\sum_{\ell=1}^{11} (M_\ell/M) \hat{\mu}_\ell^t = 0$, where M_ℓ is the number of products in group ℓ .

The estimated prior distributions differ across the network-defined groups in both their means and their dispersions. The user-group prior means range from -0.1709 in Group 3 to 0.1444 in

Group 1, a difference of 0.3154 rating points. Conditional on the product effects and under the adopted normalization, this indicates that the estimated average rating tendency is higher for users in Group 1 than for users in Group 3. The estimated within-group dispersion also varies: the prior standard deviation is 0.1580 for Group 1 and 0.6919 for Group 3. Thus, the estimated prior for Group 1 is more concentrated around its mean, whereas the prior for Group 3 allows substantially more variation among users.

Product groups show similar differences. Their prior means range from -0.1789 in Group 8 to 0.1207 in Group 4, a difference of 0.2996 rating points. Group 4 also has a relatively small prior standard deviation of 0.0945, compared with 0.3819 and 0.3864 in Groups 5 and 8. These estimates suggest that product groups differ not only in their average conditional rating components but also in how much products vary around those averages.

The group assignments are estimated from the binary matching network, rather than from the numerical ratings. Their association with differences in estimated prior means and dispersions suggests that matching patterns contain information about rating heterogeneity. In the empirical Bayes procedure, each group’s prior mean supplies a shrinkage target, while its prior variance helps determine the degree of pooling. A smaller prior variance generally implies stronger shrinkage toward the group mean, although the posterior estimate also depends on each node’s observations and its position in the bipartite network. In addition, we also evaluate held-out rating prediction using temporal and random edge splits. The result indicates that network-group EB improves on two-way effects estimation. We report them in Appendix [SM.8](#).

7 Conclusion

This paper develops an empirical Bayes procedure for estimating two-way heterogeneity when individual effects follow different distributions across groups. The procedure estimates groups on the network, fits group-specific Gaussian prior distributions, and uses the resulting posterior means to estimate individual effects. Group membership determines a distribution, not a common effect: effects may vary within each group. A directed acyclic graph summarizes the conditional dependence assumptions underlying the model. Under the stated Gaussian, group-recovery, and likelihood conditions, the estimated prior parameters are consistent, and the feasible estimator approaches the oracle posterior mean in normalized average squared distance. This result allows a vanishing fraction of nodes to be misclassified. The Monte Carlo

results show that network-informed grouping can reduce mean squared error relative to pooled and degree-group empirical Bayes when prior distributions differ across groups and the network provides sufficient information to recover those groups.

Several questions remain for future work. The theory treats group counts as fixed. Section 4 gives matrix-level sufficient conditions for likelihood concentration and stability to estimated memberships, but does not derive these conditions from a specific network-generating law. Establishing primitive network conditions for a chosen sampling model, and proving consistency of the group-count selection procedure, would strengthen the results. Further extensions could allow non-Gaussian group priors or dependence between within-group effect deviations and matching behavior. Overall, the results suggest that network-informed shrinkage is most useful when matching patterns reveal groups with meaningfully different effect distributions.

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Appendix A. Proof

Proof of Proposition 1. Connectedness implies

$$\ker(B) = \text{span}\{v_0\}.$$

Therefore, $B(a_1 - a_2) = 0$ implies

$$a_1 - a_2 = dv_0$$

for some $d \in \mathbb{R}$. Because $c'a_1 = c'a_2 = 0$ and $c'v_0 = -1$,

$$0 = c'(a_1 - a_2) = -d.$$

Thus $d = 0$, and $a_1 = a_2$. □

Proof of Proposition 2. Consider the S_n side with $K \geq 2$; the other side is analogous. Choose an orthogonal matrix attaining $e_{s,n}$, and write $F = F_{s0}O$, $\hat{F} = \hat{F}_s$, and $\hat{C} = \hat{Z}_s\hat{C}_s$. The true grouping and rotated population centers form an admissible candidate in the clustering objective. Therefore,

$$\|\hat{C} - F\|_F^2 \leq 2\|\hat{C} - \hat{F}\|_F^2 + 2\|\hat{F} - F\|_F^2 \leq 2(\kappa + 1)e_{s,n}^2.$$

Call a node bad if its fitted center is at distance at least $\delta_{s,n}/2$ from its true rotated center. The number b_n of bad nodes obeys

$$b_n \leq \frac{4\|\hat{C} - F\|_F^2}{\delta_{s,n}^2} \leq \frac{8(\kappa + 1)e_{s,n}^2}{\delta_{s,n}^2} = o_p(N).$$

By group growth, with probability tending to one every true group contains more than b_n nodes, and hence contains a good node. A fitted center cannot be within strict distance $\delta_{s,n}/2$ of two different population centers. There are exactly K fitted group labels. Consequently, on this event, the fitted centers matching good nodes establish a bijection with the K true groups. After this label alignment, every good node is correctly classified. Thus $r_{s,n} \leq b_n/N$, proving the claimed bound and consistency. When $K = 1$, the classification error is identically zero. Applying the same argument on the other side proves the result for $r_{t,n}$. Finally, $r_n \leq r_{s,n} + r_{t,n} \rightarrow_p 0$. □

Proof of Proposition 4. By Assumption 9 and the triangle inequality,

$$\sup_{\xi \in \Xi} \frac{|\Delta_n(\xi; \widehat{Z}) + \mathcal{K}_n(\xi)|}{a_n} = o_p(1).$$

Fix $\epsilon > 0$. If the parameter set outside its neighborhood is nonempty, the separation condition gives

$$\sup_{\xi \in \Xi: \|\xi - \xi_0\| \geq \epsilon} \frac{\Delta_n(\xi; \widehat{Z})}{a_n} \leq -c_\epsilon + o_p(1)$$

on events whose probabilities tend to one. In contrast, approximate maximization and feasibility of ξ_0 imply

$$\frac{\Delta_n(\widehat{\xi}; \widehat{Z})}{a_n} \geq -o_p(1).$$

Therefore $\Pr(\|\widehat{\xi} - \xi_0\| \geq \epsilon) \rightarrow 0$. If the outside set is empty, this conclusion is immediate.

For aligned product memberships,

$$\sum_{\ell=1}^L |w_\ell(\widehat{Z}) - w_\ell(Z_0)| \leq 2r_{t,n}.$$

The denominator in (12) is bounded away from zero with probability tending to one. On that event the mapping from free parameters and group shares to the full parameters is uniformly Lipschitz on the relevant compact set. Hence, for a constant C ,

$$\|\widehat{\psi} - \psi_0\| \leq C (\|\widehat{\xi} - \xi_0\| + r_{t,n}) \quad (38)$$

with probability tending to one. This proves consistency of the full hyperparameter vector. $\square$

Proof of Lemma 1. Work after label alignment and on the events where the stated bounds hold.

Their probabilities tend to one. Let

$$d_n = \|\widehat{\psi} - \psi_0\|, \quad u_0 = \alpha^{EB,O} - m_0,$$

and let S_n be the set of misclassified nodes. Then $|S_n| = p_n r_n$. Constants denoted by C below are finite and independent of the dimensions and the network.

The posterior normal equations give

$$H_0 \alpha^{EB,O} = B' Y_0 + \Lambda_0 m_0, \quad \widehat{H} \widehat{\alpha}^{EB} = B' Y_0 + \widehat{\Lambda} \widehat{m}.$$

Subtracting establishes (26). The variance bounds imply

$$\|\hat{H}^{-1}\|_{\text{op}} \leq C, \quad \|\hat{\Lambda}\|_{\text{op}} \leq C,$$

because $B'B$ is positive semidefinite and the eigenvalues of $\hat{\Lambda}$ are bounded above and away from zero.

Prior-mean error. Since the full means are bounded, a correctly classified node has prior-mean error bounded by d_n , while the error from using the wrong true group mean is bounded by a constant. Therefore,

$$\frac{\|\hat{m} - m_0\|^2}{p_n} \leq C(d_n^2 + r_n). \quad (39)$$

Prior-precision error. For this decomposition only, evaluate the full true hyperparameter vector at the estimated memberships and define

$$\bar{\Lambda} = \sigma_{\varepsilon,0}^2 \text{diag}(\hat{Z}\tau_{g0}^2)^{-1}, \quad E_n = \hat{\Lambda} - \bar{\Lambda}, \quad J_n = \bar{\Lambda} - \Lambda_0.$$

The variance bounds and fixed parameter dimension imply

$$\|E_n\|_{\text{op}} \leq Cd_n.$$

The matrix J_n is diagonal, vanishes outside S_n , and has uniformly bounded entries. Hence

$$\frac{\|(\hat{\Lambda} - \Lambda_0)u_0\|^2}{p_n} \leq Cd_n^2 \frac{\|u_0\|^2}{p_n} + \frac{C}{p_n} \sum_{i \in S_n} u_{0i}^2. \quad (40)$$

Conditional second moments. Let $\mathcal{F}_n = \sigma(B, Z_0, U_n)$, with conditioning on X understood. By the seed-independence condition, the specified conditional Gaussian model remains valid given $\mathcal{F}_n$. The oracle satisfies

$$\mathbb{E}[u_0 \mid \mathcal{F}_n] = 0,$$

$$\text{Var}(u_0 \mid \mathcal{F}_n) = D_0 B' \Omega_0^{-1} B D_0 \preceq D_0.$$

The last inequality follows from positive semidefiniteness of the Gaussian posterior covariance $D_0 - D_0 B' \Omega_0^{-1} B D_0$. Consequently,

$$\mathbb{E} \left[\frac{\|u_0\|^2}{p_n} \mid \mathcal{F}_n \right] \leq C, \quad \frac{\|u_0\|^2}{p_n} = O_p(1).$$

Because the misclassified set is $\mathcal{F}_n$ -measurable,

$$\mathbb{E} \left[\frac{1}{p_n} \sum_{i \in S_n} u_{0i}^2 \middle| \mathcal{F}_n \right] \leq C r_n.$$

Define

$$W_n = \begin{cases} |S_n|^{-1} \sum_{i \in S_n} u_{0i}^2, & |S_n| > 0, \\ 0, & |S_n| = 0. \end{cases}$$

Conditional Markov's inequality gives $W_n = O_p(1)$, and $p_n^{-1} \sum_{i \in S_n} u_{0i}^2 = r_n W_n$.

Combining (26), (39), and (40), we obtain

$$\frac{\|\widehat{\alpha}^{EB} - \alpha^{EB,O}\|^2}{p_n} \leq C \left\{ d_n^2 + r_n + d_n^2 \frac{\|u_0\|^2}{p_n} + r_n W_n \right\}.$$

This establishes (27). In particular, the random-rate notation means that the left-hand side is bounded by $d_n^2 + r_n$ times a tight nonnegative random sequence; if both rates are zero, the posterior means coincide. The proof does not condition on $\widehat{\psi}$ or require its independence from the outcome. $\square$

Proof of Theorem 1. Proposition 2 gives $r_{s,n}, r_{t,n}, r_n \rightarrow_p 0$. Proposition 4 gives $\widehat{\xi} \rightarrow_p \xi_0$ and $\|\widehat{\psi} - \psi_0\| \rightarrow_p 0$. The compactness and group-share bounds ensure that the full estimated means satisfy the boundedness required by Lemma 1. Applying that lemma gives (28). Combining the lemma with (38) gives (29). $\square$

Proof of Corollary 1. All membership labels are aligned as in Section 4.2. Conditioning on X is understood. Constants denoted by C are finite and do not depend on the dimensions or the network. By Proposition 2, the classification proportions vanish on both sides. Group growth therefore implies that the true and estimated group shares are bounded away from zero with probability tending to one. Work on these events throughout. The common free parameterization and compactness then give uniformly bounded full prior means for both Z_0 and $\widehat{Z}$.

Step 1: Uniform control of the likelihood perturbation. For a candidate ξ , abbreviate

$$m_\xi = m(\xi, \widehat{Z}), \quad \Omega_\xi = \Omega(\xi, \widehat{Z}), \quad q_\xi = Y_0 - B m_\xi.$$

Direct expansion gives the exact identity

$$\begin{aligned} \ell_n(\xi; \widehat{Z}, \widehat{Y}) - \ell_n(\xi; \widehat{Z}, Y_0) \\ = e_n' \Omega_\xi^{-1} q_\xi - \frac{1}{2} e_n' \Omega_\xi^{-1} e_n. \end{aligned} \quad (41)$$

We first establish a uniform bound on $\|\Omega_\xi^{-1} q_\xi\|$ without conditioning on the preliminary estimator.

Let $z_n = Y_0 - Bm_0$. Under the conditional Gaussian model,

$$z_n \mid B, Z_0, X \sim \mathcal{N}(0, \Omega_0), \quad z_n' \Omega_0^{-1} z_n = O_p(n).$$

The positive upper and lower variance bounds imply, uniformly in ξ and the candidate memberships,

$$\Omega_\xi \succeq c\Omega_0, \quad \Omega_\xi \succeq \underline{\sigma}^2 I_n$$

for some fixed $c > 0$. Consequently,

$$\Omega_\xi^{-2} \preceq (\underline{\sigma}^2)^{-1} \Omega_\xi^{-1} \preceq C\Omega_0^{-1}, \quad \sup_{\xi \in \Xi} \|\Omega_\xi^{-1} z_n\| = O_p(\sqrt{n}).$$

Each row of the bipartite incidence matrix B contains two unit entries. Uniform boundedness of the node prior means therefore gives

$$\sup_{\xi \in \Xi} \|B(m_0 - m_\xi)\| \leq C\sqrt{n}.$$

Since $q_\xi = z_n + B(m_0 - m_\xi)$ and $\|\Omega_\xi^{-1}\|_{\text{op}} \leq (\underline{\sigma}^2)^{-1}$, it follows that

$$\sup_{\xi \in \Xi} \|\Omega_\xi^{-1} q_\xi\| = O_p(\sqrt{n}).$$

Cauchy–Schwarz applied to (41) yields

$$\begin{aligned} A_n &:= \sup_{\xi \in \Xi} |\ell_n(\xi; \widehat{Z}, \widehat{Y}) - \ell_n(\xi; \widehat{Z}, Y_0)| \\ &\leq \|e_n\| \sup_{\xi \in \Xi} \|\Omega_\xi^{-1} q_\xi\| + C\|e_n\|^2 = O_p(\sqrt{n}). \end{aligned} \quad (42)$$

The last equality uses $\|e_n\| = O_p(1)$. In particular, $A_n/a_n \rightarrow_p 0$ by Assumption 11.

Step 2: Consistency of the feasible hyperparameters. Define the feasible likelihood contrast

$$\tilde{\Delta}_n(\xi) = \ell_n(\xi; \hat{Z}, \hat{Y}) - \ell_n(\xi_0; \hat{Z}, \hat{Y}).$$

The contrast $\Delta_n(\xi; \hat{Z})$ from Section 4.3 uses Y_0 in both terms. Step 1 implies

$$\sup_{\xi \in \Xi} |\tilde{\Delta}_n(\xi) - \Delta_n(\xi; \hat{Z})| \leq 2A_n = o_p(a_n).$$

Combining this bound with the concentration and classification-stability parts of Assumption 9 gives

$$\sup_{\xi \in \Xi} \frac{|\tilde{\Delta}_n(\xi) + \mathcal{K}_n(\xi)|}{a_n} \xrightarrow{p} 0.$$

For every fixed $\epsilon > 0$ with a nonempty outside set, the separation condition therefore implies

$$\sup_{\xi \in \Xi: \|\xi - \xi_0\| \geq \epsilon} \frac{\tilde{\Delta}_n(\xi)}{a_n} \leq -c_\epsilon + o_p(1)$$

on events whose probabilities tend to one. Approximate maximization in (31), by contrast, gives $\tilde{\Delta}_n(\tilde{\xi})/a_n \geq -o_p(1)$. Thus $\tilde{\xi} \rightarrow_p \xi_0$. The same group-share argument used in Proposition 4 gives, with probability tending to one,

$$\|\tilde{\psi} - \psi_0\| \leq C(\|\tilde{\xi} - \xi_0\| + r_{t,n}) \xrightarrow{p} 0. \quad (43)$$

Step 3: Direct effect of residualization on the posterior mean. For any candidate define

$$\Lambda = \sigma_\varepsilon^2 D(\xi, Z)^{-1}, \quad H = B'B + \Lambda, \quad \lambda_* := \frac{\sigma^2}{\tau^2} > 0.$$

The posterior updating matrix satisfies

$$D(\xi, Z)B'\Omega(\xi, Z)^{-1} = H^{-1}B', \quad \Lambda \succeq \lambda_* I_{p_n}.$$

Moreover,

$$(H^{-1}B')(H^{-1}B')' = H^{-1}B'BH^{-1} = H^{-1} - H^{-1}\Lambda H^{-1} \preceq H^{-1} \preceq \lambda_*^{-1} I_{p_n}.$$

Hence $\|H^{-1}B'\|_{\text{op}} \leq \lambda_*^{-1/2}$ uniformly. Introduce the intermediate predictor

$$\alpha^* = T_n(Y_0; \tilde{\xi}, \hat{Z}).$$

It uses the hyperparameters estimated from $\hat{Y}$, but evaluates the posterior map at Y_0 . By linearity in the outcome,

$$\tilde{\alpha}^{EB} - \alpha^* = -\tilde{H}^{-1}B'e_n, \quad \frac{\|\tilde{\alpha}^{EB} - \alpha^*\|^2}{p_n} \leq \frac{\|e_n\|^2}{\lambda_* p_n} = O_p(p_n^{-1}).$$

Step 4: Combine posterior perturbations. Lemma 1 applies to α^* . It allows outcome-dependent estimated hyperparameters and therefore does not require independence of $\tilde{\psi}$ from Y_0 . The memberships still use only the network and the permitted seed. Consequently,

$$\frac{\|\alpha^* - \alpha^{EB,O}\|^2}{p_n} = O_p(r_n + \|\tilde{\psi} - \psi_0\|^2).$$

Using $\|u + v\|^2 \leq 2\|u\|^2 + 2\|v\|^2$, Step 3, and (43), we obtain

$$\frac{\|\tilde{\alpha}^{EB} - \alpha^{EB,O}\|^2}{p_n} = O_p(r_n + \|\tilde{\xi} - \xi_0\|^2 + r_{t,n}^2 + p_n^{-1}) = o_p(1).$$

This proves both assertions concerning posterior-mean approximation. For the more general preliminary estimators in the remark, retain $\|e_n\|$ in Steps 1 and 3. The two displayed conditions in that remark give the same consistency conclusions. $\square$

Proof of Proposition 3. All calculations below are conditional on (B, Z_0) . By Assumption 6, the estimated memberships are also fixed under this conditioning, apart from algorithmic randomness independent of the outcome disturbance.

For the oracle likelihood, the Gaussian log-likelihood formula gives (18). Since z_n is standard normal and C_ξ is symmetric,

$$\text{Var}(z_n' C_\xi z_n) = 2 \text{tr}(C_\xi^2), \quad \text{Var}(v_\xi' z_n) = \|v_\xi\|^2.$$

The covariance between the centered quadratic and linear terms is zero because centered Gaussian vectors have zero third moments. This proves (19). The Gaussian quadratic-form tail bound,

together with (21), gives for each fixed ξ and every $\epsilon > 0$

$$\Pr(|\Delta_n(\xi; Z_0) + \mathcal{K}_n(\xi)| > \epsilon a_n \mid B, Z_0) \leq 2 \exp(-c_\epsilon a_n)$$

on events whose probabilities tend to one. Cover the compact, fixed-dimensional Ξ by a deterministic net of mesh $a_n^{-1/4}$. Its cardinality is polynomial in a_n , so the union bound makes the maximum over the net $o_p(a_n)$. The assumed gradient bound makes the interpolation error at most $a_n^{-1/4} O_p(a_n) = o_p(a_n)$. This proves part (ii).

For the membership perturbation, put

$$R_\xi = (I_n + Q_\xi)^{-1} - I_n, \quad g_\xi = \Omega_\xi^{-1/2}(Y_0 - b_0) - s_\xi.$$

The random vector g_ξ is conditionally Gaussian with covariance $\Omega_\xi^{-1/2} \Omega_0 \Omega_\xi^{-1/2}$, whose operator norm is uniformly bounded by the variance bounds in Assumption 7. The eigenvalue bounds in (22) imply

$$\|R_\xi\|_{\text{op}} \leq C \|Q_\xi\|_{\text{op}}, \quad \|R_\xi\|_* = O(\|Q_\xi\|_*), \quad \|R_\xi\|_F^2 \leq \|R_\xi\|_{\text{op}} \|R_\xi\|_* = o_p(a_n)$$

uniformly over Ξ . Direct expansion of the two Gaussian log-likelihoods yields

$$\begin{aligned} \ell_n(\xi; \widehat{Z}) - \ell_n(\xi; Z_0) &= -\frac{1}{2} \log |I_n + Q_\xi| - \frac{1}{2} (s_\xi + g_\xi)' R_\xi (s_\xi + g_\xi) \\ &\quad + u'_\xi (I_n + Q_\xi)^{-1} (s_\xi + g_\xi) - \frac{1}{2} u'_\xi (I_n + Q_\xi)^{-1} u_\xi. \end{aligned}$$

The log-determinant term is bounded by $C \|Q_\xi\|_*$. The deterministic quadratic term involving s_ξ is $o_p(a_n)$, since $\|Q_\xi\|_{\text{op}} = o_p(1)$ and $\|s_\xi\|^2 = O_p(a_n)$. The centered Gaussian quadratic term has variance bounded by $C \|R_\xi\|_F^2 = o_p(a_n)$; its linear component has variance bounded by $C \|R_\xi\|_{\text{op}}^2 \|s_\xi\|^2 = o_p(a_n)$. The terms involving u_ξ are $o_p(a_n)$, because $\|u_\xi\|^2 = o_p(a_n)$ and $\|u_\xi\| \|s_\xi\| = o_p(a_n)$. Thus the likelihood perturbation is $o_p(a_n)$ at every point of the net. The same conditional Gaussian tail bounds, the net cardinality above, and the assumed gradient bound extend this result uniformly over Ξ . Subtracting the perturbation at ξ_0 from that at ξ changes the bound by at most a factor of two. This proves part (iii), completing the proof. $\square$

Appendix B. Supplementary Material

SM.1 Derive Working Posterior from Equation (4)

To derive this estimator, we begin with the joint distribution implied by the graphical model in Figure 2. Omitting the hyperparameters from the notation for brevity, the joint distribution in equation (3) can be decomposed further as follows:

$$\begin{aligned}\mathbb{P}(\alpha \mid \tilde{Y}, B) &\propto \sum_Z \mathbb{P}(\tilde{Y}, B, Z, \alpha) \\ &= \sum_Z \mathbb{P}(\tilde{Y} \mid \alpha, B) \mathbb{P}(B \mid Z) \mathbb{P}(\alpha \mid Z) \mathbb{P}(Z)\end{aligned}\tag{44}$$

$$\propto \sum_Z \mathbb{P}(\tilde{Y} \mid \alpha, B) \mathbb{P}(Z \mid B) \mathbb{P}(\alpha \mid Z),\tag{45}$$

$$\begin{aligned}&\text{conditional on estimating } Z \text{ such that } \mathbb{P}(\hat{Z} \mid B) \rightarrow 1, \\ &\rightarrow \mathbb{P}(\tilde{Y} \mid \alpha, B) \mathbb{P}(\alpha \mid \hat{Z}).\end{aligned}\tag{46}$$

We suppose that the latent group structure can be consistently recovered from the observed network B , in the sense that

$$\mathbb{P}(\hat{Z} = Z \mid B) \rightarrow 1.$$

Under this condition, posterior uncertainty over the group memberships vanishes asymptotically, and equation (45) reduces to the feasible representation in equation (46), with the unknown Z replaced by $\hat{Z}$. If the parameters in prior are learned from data, then the posterior mean is the empirical Bayes estimator.

SM.2 ML-EB Estimator

We discuss the estimation in detail. The covariance matrix of α conditional on Z is

$$D(Z\tau_g^2) = \text{diag}(Z\tau_g^2).$$

Consequently,

$$\alpha \mid Z \sim \mathcal{N}\left(Z\mu_g, D(Z\tau_g^2)\right).$$

For the Gaussian benchmark, assume

$$\varepsilon \mid (B, X) \sim \mathcal{N}(0, \sigma_\varepsilon^2 I_n).$$

When β_0 is known, it follows that the marginal distribution of $\tilde{Y} = Y - X\beta_0$ is

$$\tilde{Y} \mid B, Z \sim \mathcal{N}\left(BZ\mu_g, BD(Z\tau_g^2)B' + \sigma_\varepsilon^2 I_n\right). \quad (47)$$

Equation (47) provides the basis for both identification of the group-specific hyperparameters and their marginal maximum-likelihood estimation.

For a candidate membership matrix and hyperparameters, the Gaussian model implies the following joint normal distribution. When memberships are estimated, this is the plug-in working model conditional on $\hat{Z}$.

$$\begin{bmatrix} \tilde{Y} \\ \alpha \end{bmatrix} \mid B, \hat{Z} \sim \mathcal{N}\left(\begin{bmatrix} B\hat{Z}\mu_g \\ \hat{Z}\mu_g \end{bmatrix}, \begin{bmatrix} BD(\hat{Z}\tau_g^2)B' + \sigma_\varepsilon^2 I_n & BD(\hat{Z}\tau_g^2) \\ D(\hat{Z}\tau_g^2)B' & D(\hat{Z}\tau_g^2) \end{bmatrix}\right). \quad (48)$$

Here the plug-in prior mean is $\hat{Z}\hat{\mu}_g$, and $\hat{D} = \text{diag}(\hat{Z}\hat{\tau}_g^2)$ is the diagonal matrix of estimated node-level prior variances, with parameters defined after estimating the marginal likelihood.

The conditional Gaussian posterior-mean map has the following closed form. In the known- β_0 case it uses $\tilde{Y}$; in the feasible estimated-coefficient procedure, the same map is evaluated at $\hat{Y}$, as described in Section 4.5.

$$\hat{Z}\hat{\mu}_g + \hat{D}B' (B\hat{D}B' + \hat{\sigma}_\varepsilon^2 I_n)^{-1} (\tilde{Y} - B\hat{Z}\hat{\mu}_g), \quad (49)$$

Given $\hat{Z}$, the group-level means, variances, and disturbance variance are estimated by marginal maximum likelihood using equation (50); their estimates are then substituted in equation (49) to obtain the posterior mean.

$$\mathbb{P}(\tilde{Y} \mid B, \hat{Z}, \psi) = \int_{\alpha} \mathbb{P}(\tilde{Y} \mid \alpha, B) \mathbb{P}(\alpha \mid \hat{Z}, \psi) d\alpha, \quad (50)$$

which give us the estimated parameters $\hat{\psi} = (\mu_g, \tau_g^2, \sigma_\varepsilon^2)$ as follows,

$$\hat{\psi}^{MLE} = \arg \max_{\psi \in \mathcal{J}} \mathbb{P}(\tilde{Y} \mid B, \hat{Z}, \psi). \quad (51)$$

Substituting the marginal maximum likelihood estimates $\hat{\psi}^{MLE}$ into equation (49) yields the empirical Bayes estimator. We summarize the estimation procedure in the following algorithm.

SM.3 Additional Material for Simulation

SM.3.1 Evaluation criteria and the value of network information

Mean squared error criteria. The primary criterion is pair-sum mean squared error:

$$\text{MSE}_{\text{pair}} = \frac{1}{NM} \sum_{i,j} \left\{ (\hat{\alpha}_i^s + \hat{\alpha}_j^t) - (\alpha_i^s + \alpha_j^t) \right\}^2. \quad (52)$$

It evaluates the estimated two-way effect contribution uniformly over all user-product pairs, including those without an observed link, and is invariant to the unidentified common location shift between the two sides. For each estimator, we also compute the raw-effect mean squared error

$$\text{MSE}_{\text{raw}} = \frac{1}{N+M} \left\{ \sum_i (\hat{\alpha}_i^s - \alpha_i^s)^2 + \sum_j (\hat{\alpha}_j^t - \alpha_j^t)^2 \right\}. \quad (53)$$

This mean squared error evaluates prediction of the underlying random effects in the hierarchical model. However, the outcome likelihood alone does not identify the allocation of a common location shift between sides. For OLS, the computational convention will be $\sum_j \hat{\alpha}_j^t = 0$. Under the simulated independent prior, the realized $\sum_j \alpha_j^t$ need not be zero. Accordingly, raw OLS mean squared error includes the consequences of this location convention and will not be interpreted as error for an unbiased predictor of the unnormalized effect vector.

As a secondary criterion, we evaluate identified within-side heterogeneity: write $\bar{\alpha}^s = N^{-1} \sum_i \alpha_i^s$, $\bar{\hat{\alpha}}^s = N^{-1} \sum_i \hat{\alpha}_i^s$, and define the corresponding means on side t analogously.

$$\begin{aligned} \text{MSE}_s^c &= \frac{1}{N} \sum_i \left\{ (\hat{\alpha}_i^s - \bar{\hat{\alpha}}^s) - (\alpha_i^s - \bar{\alpha}^s) \right\}^2, \\ \text{MSE}_t^c &= \frac{1}{M} \sum_j \left\{ (\hat{\alpha}_j^t - \bar{\hat{\alpha}}^t) - (\alpha_j^t - \bar{\alpha}^t) \right\}^2, \end{aligned} \quad (54)$$

and $\text{MSE}_c = (N \text{MSE}_s^c + M \text{MSE}_t^c) / (N + M)$. Centering is used only to define this secondary evaluation target; it does not change the generated effects or the fitted hierarchical model. The program also records centered MSE separately for each side. It does not compute degree-bin-specific MSEs, so none are reported here.

Writing $a_i = \hat{\alpha}_i^s - \alpha_i^s$ and $b_j = \hat{\alpha}_j^t - \alpha_j^t$, it can be computed as

$$\text{MSE}_{\text{pair}} = \frac{1}{N} \sum_i a_i^2 + \frac{1}{M} \sum_j b_j^2 + 2\bar{a}\bar{b}.$$

An additional observed-edge mean squared error averages the same squared effect-sum errors over $(i, j) \in \mathcal{E}_n$. Unlike (52), it weights nodes through their realized matching frequencies. These are conditional-mean prediction MSEs; they do not include fresh outcome noise. In the estimated-regression experiment, complete outcome-mean prediction will additionally include $x_{ij}(\hat{\beta} - \beta_0)$, evaluated on independent test covariates with the specified distribution.

Incremental value of learning groups. The central comparison is network-group EB against ordinary EB. Degree-group EB assesses whether detailed matching patterns add information beyond activity. Comparisons across the four prior configurations will distinguish successful community detection from useful prior information. In particular, no improvement is presumed in the configuration with identical effect distributions across groups.

Comparisons with true-group EB, known-prior EB, and the full oracle will separately examine membership learning, hyperparameter estimation, and residualization. These comparisons need not yield nonnegative realized MSE differences in each replication. Under the correctly specified Gaussian model, oracle approximation will also be measured by

$$Q_n = \frac{1}{N + M} \|\hat{\alpha}^{EB} - \alpha^{EB,O}\|^2, \quad (55)$$

using the feasible predictor from the relevant known- or estimated-coefficient experiment.

SM.3.2 Classification error, sparse observations, and weak connectivity

Classification accuracy and controlled errors. Classification proportions will be reported separately on each side after optimal label alignment:

$$r_s = \min_{\Pi_s} \frac{\|\hat{Z}_s - Z_{s0}\Pi_s\|_F^2}{2N}, \quad r_t = \min_{\Pi_t} \frac{\|\hat{Z}_t - Z_{t0}\Pi_t\|_F^2}{2M}.$$

An additional experiment will begin with the true memberships and deliberately corrupt approximately $\rho \in \{0, 0.01, 0.05, 0.10, 0.20\}$ of the labels on each side. Balanced swaps will preserve group sizes, and corrupted nodes will be selected independently of effect innovations. Realized

error proportions after optimal alignment will be recorded.

For each corrupted partition, the posterior will first be evaluated using the true full hyperparameters assigned through that partition, and then evaluated after re-estimating the hyperparameters. The first comparison isolates direct sensitivity of posterior updating to classification error; the second includes its effect on prior estimation. The relationship between Q_n and the realized error proportions will be examined without presuming replication-by-replication monotonicity.

Weak connectivity at approximately fixed internal density. To separate weak global connectivity from low degree, a supplementary design will use $K = L = 2$, with equal group sizes on each side. Within each corresponding pair of groups, generate a connected bipartite graph with target average internal degree 24. Then add exactly $b \in \{1, 3, 10, 30\}$ distinct edges, sampled uniformly from the two off-diagonal blocks. No other cross-community edges are included. Use group means $(-0.6, 0.6)$, variances $(0.09, 0.09)$ on each side, and the same covariate and disturbance specification as the baseline. Effects remain independent of link formation conditional on memberships.

For each graph, report the smallest positive eigenvalue of $B'B$ and the second-smallest eigenvalue of the normalized Laplacian of the full undirected bipartite graph. This design examines settings in which groups may be readily detected while their relative locations are informed by few cross-community observations. Fixed small values of b are stress tests, not presumed examples satisfying the information-growth assumptions for hyperparameter consistency.

Increasing network size. Selected cells will also be evaluated at $N = M \in \{600, 1200, 2400\}$, using the same group proportions. A fixed-degree sequence will describe finite-information behavior. A separate increasing-degree sequence, $d_N = 24 \log(N) / \log(600)$, will examine whether classification errors and oracle distances decrease as information grows. These experiments provide numerical diagnostics, rather than verification of all asymptotic likelihood conditions.

SM.3.3 Robustness and simulation reporting

Departures from the Gaussian group model. Robustness experiments will use selected baseline cells and change one feature at a time. First, the effect innovations will be replaced by $\sqrt{3/5} t_5$ variables, preserving unit variance while introducing heavier tails. Second, let g_i deter-

mine network formation but generate a separate effect label c_i : with probability $1 - q$, set $c_i = g_i$; otherwise draw c_i uniformly from the three labels. Apply the same mechanism independently on the other side, with $q \in \{0, 0.25, 0.50, 1\}$. Generate effects from the distributions indexed by the effect labels. At $q = 1$, the network labels are uninformative about the effect labels. For intermediate q , effects conditional on network labels follow mixtures rather than the single Gaussian distributions assumed by the estimator.

Third, a continuous-type design will draw independent $v_i^s, v_j^t \sim \text{Uniform}[-1, 1]$, generate effects

$$\alpha_i^s = 0.6v_i^s + 0.3u_i^s, \quad \alpha_j^t = 0.6v_j^t + 0.3u_j^t,$$

and generate links with probabilities

$$P_{ij} = \text{logit}^{-1}\{a + 2v_i^s v_j^t + \log \theta_i + \log \phi_j\}.$$

Conditional on the types and propensities, choose a numerically to give $\sum_{i,j} P_{ij} = Nd$ before connectedness conditioning. This design has no finite true partition; fitting three groups per side evaluates a discrete approximation to continuous heterogeneity.

In misspecified designs, the Gaussian predictor supplied with the actual conditional first two moments, when available, will be called a known-moment Gaussian benchmark. It will not be called the true posterior mean. Benchmarks given otherwise unobserved effect labels will be identified as possessing additional information. No Bayes optimality claim for a working Gaussian benchmark will be made.

Degree heterogeneity, imbalance, and group counts. The baseline propensity distribution will be compared with constant propensities and a more dispersed bounded distribution, with mean one within each group. Every design will check that edge probabilities are valid; probabilities will not be silently clipped. Connectedness acceptance rates and low-degree-node coverage will be reported. If a largest-component analysis is added, it will be identified separately, with its retained population and group composition documented.

Additional experiments will consider group shares $(0.2, 0.3, 0.5)$ and a rectangular design with $N = 600$, $M = 900$, $K = 2$, and $L = 3$. For the rectangular design, use the block-affinity

matrix

$$W = \begin{pmatrix} 3 & 1 & 2 \\ 1 & 3 & 2 \end{pmatrix}, \quad P_{ij} = c\theta_i\phi_jW_{g_ih_j}, \quad c = \frac{Nd}{\sum_{i,j}\theta_i\phi_jW_{g_ih_j}},$$

with $d = 24$, equal group shares within each side, and the baseline propensity distribution. Use means $(-0.6, 0.6)$ and $(-0.6, 0, 0.6)$ on the respective sides and common variance 0.09. The spectral implementation will retain two leading singular vectors and form two and three clusters from the corresponding left and right embeddings. Population row directions must be checked for separation; a one-dimensional unit-normalized embedding will not be used to identify three groups.

The main results condition on known numbers of groups. A separate experiment will use the implemented network-only selection criterion, with its exact penalty, search grid, and optimization procedure documented. It will report selected-count frequencies and estimation MSE both unconditionally and conditional on correct count selection. No consistency claim follows merely from using a modified information criterion. One-group cases require explicit handling rather than an empty spectral embedding.

Monte Carlo precision and reproducibility. The reported main study uses 200 independent replications per cell, with the two regression versions evaluated on the same generated samples. Membership assignments, network propensities, graphs, effects, covariates, and disturbances will be redrawn across replications. Within each replication, all estimators use the same generated data. Random seeds, software versions, design parameters, and replication-level results will be saved. Numerical checks will include posterior formulas on small graphs, likelihood evaluations, label alignment, and the MSE definitions.

For the primary pair-sum mean squared error criterion, the principal comparison uses paired differences

$$D_r = \text{MSE}_{\text{pair,ordinary EB},r} - \text{MSE}_{\text{pair,network EB},r}, \quad \bar{D} = \frac{1}{R} \sum_{r=1}^R D_r,$$

with Monte Carlo standard error

$$\text{MCSE}(\bar{D}) = \frac{\text{sd}(D_1, \dots, D_R)}{\sqrt{R}}.$$

Positive $\bar{D}$ favors network-group EB. The workbook also reports centered within-side MSE, raw-

effect MSE, observed-edge mean squared error, test mean squared error, and oracle distance. Dispersion of parameter estimates across replications will be labeled as a Monte Carlo standard deviation, not confused with the standard error of their average. Estimation failures will not be silently discarded; the reported comparison will state failure counts and the treatment of incomplete paired observations.

The reported main-factorial summaries average each estimator’s pair-sum MSE equally over the 36 design cells and plot paired improvement over ordinary EB against community strength. The workbook also records centered MSE and classification diagnostics. Degree-bin-specific MSE and bridge-design results are not available in the current run. Hyperparameter bias, Monte Carlo dispersion, boundary frequencies, and computation times are recorded where available; results are reported only for completed, valid designs.

SM.4 Supplementary Monte Carlo Designs and Results

This appendix reports robustness checks that complement the main factorial. Each design has 20 replications. Table 4 reports pair-sum MSE for the estimated- β_0 case. Its final row equally weights the 14 distinct supplementary configurations for which valid results are available. The average is descriptive because these configurations represent different stress tests and are not draws from a common distribution of designs.

Table 4: Pair-sum MSE in supplementary simulation designs

Design	OLS	Pooled EB	Degree-group EB	Co-spectral EB
No network signal; common prior distributions	0.0974	0.0612	0.0616	0.0615
No network signal; heterogeneous prior distributions	0.0965	0.0829	0.0831	0.0830
Degree propensities: constant	0.0896	0.0747	0.0749	0.0530
Degree propensities: high heterogeneity	0.2443	0.1416	0.1423	0.1074
Heavy-tailed effects	0.0980	0.0799	0.0801	0.0551
Network–effect group mismatch, $q = 0.25$	0.0975	0.0813	0.0814	0.0760
Network–effect group mismatch, $q = 0.50$	0.0969	0.0816	0.0817	0.0798
Continuous effect heterogeneity	0.0971	0.0725	0.0720	0.0695
Unequal group shares	0.1022	0.0825	0.0807	0.0631
Fixed degree, $N = 600$, $d = 24$	0.0973	0.0788	0.0789	0.0545
Fixed degree, $N = 1200$, $d = 24$	0.0973	0.0795	0.0796	0.0556
Increasing degree, $N = 1200$, $d \approx 26.60$	0.0875	0.0730	0.0731	0.0516
10% controlled membership misclassification	0.0967	0.0787	0.0789	0.0557
Network-only group-count selection	0.0978	0.0799	0.0801	0.0557
Equal-weight average over 14 distinct valid designs	0.1069	0.0820	0.0820	0.0658

Notes: Entries are mean pair-sum MSEs over 20 replications, using estimated β_0 . The average row excludes three designs with no valid replications and excludes the duplicated $N = 600$, $d = 24$ increasing-degree baseline.

The negative-control designs show that network grouping does not improve pair-sum mean

squared error when the network contains no community signal. With identical prior distributions, co-spectral empirical Bayes has pair-sum mean squared error 0.0615, compared with 0.0612 for pooled empirical Bayes. With heterogeneous prior distributions but no network signal, the corresponding values are 0.0830 and 0.0829. By contrast, the advantage attenuates gradually as network and effect groups become less aligned: the pair-sum MSE improvement over pooled empirical Bayes is 0.0053 at $q = 0.25$ and 0.0018 at $q = 0.50$.

The method also has lower pair-sum mean squared error than pooled empirical Bayes in the heavy-tailed, continuous-heterogeneity, unequal-share, and degree-heterogeneity designs. These results suggest that the gains are not limited to the baseline Gaussian design, although they do not establish robustness to arbitrary prior misspecification. Across the fixed-degree growth designs, co-spectral empirical Bayes has pair-sum MSE 0.0545 at $N = 600$ and 0.0556 at $N = 1200$. At $N = 1200$, increasing the expected degree to approximately 26.60 reduces pair-sum MSE to 0.0516. These finite-size results are descriptive and do not establish an asymptotic rate.

In the controlled-misclassification design, the co-spectral procedure misclassifies about 0.2% of nodes on average. When 10% of labels are deliberately corrupted, pair-sum mean squared error is 0.0891 when the prior moments are held fixed and 0.0696 when they are re-estimated. The latter adjustment reduces, but does not eliminate, the pair-sum MSE increase relative to co-spectral empirical Bayes in this design (0.0557).

The network-only group-count procedure selects $K = L = 3$ in all 200 replications. Its pair-sum mean squared error, 0.0557, is nearly identical to the 0.0557 obtained when the true counts are supplied. This result is encouraging for this design, which has clear network communities, but one design is not enough to establish general group-count consistency. The oracle estimators using true groups or true prior moments continue to have lower average pair-sum MSE than the feasible co-spectral estimator.

Simulation coverage and implementation. The supplementary suite planned 18 designs. No valid replications were produced for the rectangular $K = 2, L = 3$ design or the weak-connectivity designs with 1 and 10 bridges. In all three cases, the data-generation routine failed because the generated membership labels did not match the group counts specified by the design. These failures concern the simulation implementation, not estimator performance, and the three designs are excluded from Table 4 and its average. The nominal increasing-degree design at $N = 600$ also has $d = 24$, the same as the fixed-degree design at that size; it is therefore

treated as a duplicate baseline and excluded from the equal-weight average. The rectangular and weak-connectivity results should be added after correcting and rerunning those designs.

SM.5 Determine the Number of Groups

We first use the observed user–product interaction network to recover latent groups of users and products. Importantly, the group-detection procedure is based only on whether a user has reviewed a product and does not use the numerical rating outcome. This separation ensures that the dependent variable in the subsequent empirical Bayes analysis does not mechanically determine the latent group memberships. Let

$$A \in \{0, 1\}^{N \times M}$$

denote the rectangular bi-adjacency matrix of the bipartite network, where N is the number of users and M is the number of products. Its (i, j) -th element is defined as

$$A_{ij} = \begin{cases} 1, & \text{if user } i \text{ rated product } j, \\ 0, & \text{otherwise.} \end{cases}$$

In the empirical application, the network consists of $N = 50,985$ users, $M = 25,848$ products, and $E = 412,947$ observed user–product links. The degree of user i and the degree of product j are respectively defined as

$$d_i = \sum_{j=1}^M A_{ij}, \quad q_j = \sum_{i=1}^N A_{ij}.$$

These degrees vary substantially across nodes, reflecting pronounced heterogeneity in user activity and product popularity.

To prevent such degree heterogeneity from being mistaken for genuine community structure, we employ a degree-corrected bipartite stochastic block model. The specification is motivated by the degree-corrected stochastic block model of ?, adapted here to a network with two distinct sets of nodes. Let

$$g_i \in \{1, \dots, K\}$$

denote the latent group membership of user i , and let

$$h_j \in \{1, \dots, L\}$$

denote the latent group membership of product j . In contrast to a one-mode block model, the numbers of groups on the two sides of the network are not required to be equal. Under the

Poisson formulation of the degree-corrected model, the conditional edge intensity is

$$\mathbb{E}[A_{ij} \mid g_i, h_j] = \theta_i \phi_j \omega_{g_i h_j},$$

where θ_i captures the activity propensity of user i , ϕ_j captures the popularity propensity of product j , and ω_{kl} measures the underlying matching affinity between user group k and product group l . Thus, the expected interaction intensity can be interpreted as the product of user activity, product popularity, and group-level matching affinity.

The multiplicative representation requires a normalization because the degree parameters and block-affinity parameters are otherwise not separately identified. We impose the within-group normalizations

$$\sum_{i: g_i=k} \theta_i = 1, \quad k = 1, \dots, K,$$

and

$$\sum_{j: h_j=l} \phi_j = 1, \quad l = 1, \dots, L.$$

Define the total degree associated with user group k and product group l as

$$\kappa_k = \sum_{i: g_i=k} d_i, \quad \eta_l = \sum_{j: h_j=l} q_j,$$

and define the observed number of links between the corresponding pair of groups as

$$e_{kl} = \sum_{i: g_i=k} \sum_{j: h_j=l} A_{ij}.$$

Conditional on a given partition, the maximum-likelihood estimates of the node-specific degree parameters take the natural form

$$\hat{\theta}_i = \frac{d_i}{\kappa_{g_i}}, \quad \hat{\phi}_j = \frac{q_j}{\eta_{h_j}},$$

while, under the normalization above, the block-intensity estimator is proportional to the corresponding block-edge count e_{kl} . Consequently, heterogeneity in the overall activity of individual users and the popularity of individual products is absorbed by the node-specific parameters rather than by the latent group assignments.

Profiling out the degree parameters yields a convenient objective for evaluating a candidate

partition. Let

$$E = \sum_{i=1}^N \sum_{j=1}^M A_{ij}$$

denote the total number of observed edges. Conditional on the degree sequence, the expected number of edges between user group k and product group l under random bipartite matching is

$$e_{kl}^{(0)} = \frac{\kappa_k \eta_l}{E}.$$

We therefore evaluate a candidate partition using the degree-corrected profile objective

$$Q_{\text{DC}}(K, L) = \sum_{k=1}^K \sum_{l=1}^L e_{kl} \log \left(\frac{e_{kl}}{e_{kl}^{(0)}} \right) = \sum_{k=1}^K \sum_{l=1}^L e_{kl} \log \left(\frac{e_{kl} E}{\kappa_k \eta_l} \right),$$

where a term with $e_{kl} = 0$ is defined to be zero. The factor E inside the logarithm contributes only the constant $E \log E$ across candidate partitions and therefore does not affect the ranking of alternative group assignments. Economically, Q_{DC} measures the extent to which observed user-group-product-group interactions differ from what would be expected solely from heterogeneous user activity and product popularity.

Direct maximization of the block-model likelihood over all possible user and product assignments is computationally infeasible for a network of this size. We therefore estimate the latent memberships using degree-normalized spectral co-clustering, following the normalized asymmetric clustering framework of [Rohe et al. \(2016\)](#). Let

$$D_u = \text{diag}(d_1, \dots, d_N), \quad D_p = \text{diag}(q_1, \dots, q_M),$$

and construct the normalized bi-adjacency matrix

$$\tilde{A} = D_u^{-1/2} A D_p^{-1/2}.$$

We then compute its truncated singular value decomposition,

$$\tilde{A} = U \Sigma V^\top.$$

For a candidate pair (K, L) , let $r = \min\{K, L\}$ and retain the r leading singular directions,

including the leading direction:

$$U_r = [u_1, \dots, u_r], \quad V_r = [v_1, \dots, v_r].$$

The rows of these two matrices are normalized to unit Euclidean length whenever their norms are nonzero. We then cluster the rows of U_r into K user groups and the rows of V_r into L product groups, separately, using MiniBatch K -means. Including the leading direction avoids reducing the $(K, L) = (2, 3)$ candidate to a one-dimensional product embedding: both embeddings then have two coordinates, so three distinct product-group centers are possible, subject to the required population separation condition. The same construction also gives a nonempty embedding for $(K, L) = (1, 1)$.

Under the degree-corrected factorization above, a one-group specification on only one side does not separately identify the other side’s group-specific affinities from its node-degree parameters. Accordingly, the candidate set $\mathcal{C}$ is restricted to pairs with $K, L \geq 2$, together with the single candidate $(1, 1)$ if the no-community model is included. This restriction must be reflected in the actual search grid. In the implementation, several independent MiniBatch K -means starts are considered for each candidate pair. For a fixed (K, L) , we retain the pair of assignments with the largest value of $Q_{\text{DC}}(K, L)$.

Although the degree-corrected profile objective increases the fit of the model, it cannot by itself determine the appropriate numbers of groups, because larger values of K and L generally provide greater flexibility. We therefore introduce a complexity penalty motivated by the corrected Bayesian information criterion of [Hu et al. \(2020\)](#). Their one-mode CBIC explicitly penalizes the latent community assignments in addition to the usual model-parameter complexity. In a one-mode network with n nodes and k groups, there are approximately k^n possible membership configurations, whose logarithm is $n \log k$. In the present bipartite setting, the user side has approximately K^N possible assignments and the product side has approximately L^M possible assignments. Hence, their combined log assignment complexity is

$$N \log K + M \log L.$$

Because our network is rectangular, the block-affinity matrix contains KL parameters rather than the $k(k+1)/2$ parameters arising from a symmetric one-mode block matrix. Motivated by

this distinction, we use the following degree-corrected bipartite adaptation of the CBIC:

$$\text{CBIC}_{\text{DC}}(K, L) = -2Q_{\text{DC}}(K, L) + 2N \log K + 2M \log L + KL \log(NM).$$

The first term rewards goodness of fit, the second and third terms penalize the uncertainty associated with user and product membership assignments, respectively, and the final term penalizes the KL block-affinity parameters. We emphasize that this exact rectangular expression is a bipartite adaptation of the CBIC principle rather than a formula stated verbatim in [Hu et al. \(2020\)](#). The numbers of user and product groups are therefore selected jointly according to

$$(\widehat{K}, \widehat{L}) = \arg \min_{(K, L) \in \mathcal{C}} \text{CBIC}_{\text{DC}}(K, L).$$

In the empirical implementation, we first examine a broad set of candidate pairs in $\mathcal{C}$, and subsequently conduct a finer search around the region attaining the smallest criterion values. Multiple clustering initializations are evaluated for each candidate specification, and the partition attaining the highest degree-corrected profile objective is used to calculate the corresponding CBIC. This procedure selects

$$\widehat{K} = 10, \quad \widehat{L} = 11,$$

indicating that the latent structure of the user side and product side cannot be adequately represented by imposing an equal number of groups. Because cluster labels are arbitrary up to permutation, the estimated block structure is not required to be diagonal: for example, user group k may have its strongest interaction affinity with product group $l \neq k$. Accordingly, the economically relevant objects are the estimated group-to-group affinities rather than the numerical labels assigned to the groups.

Finally, the estimated memberships are used as inputs to the empirical Bayes estimation of heterogeneous user and product effects. Let

$$\widehat{Z}_u \in \{0, 1\}^{N \times \widehat{K}}$$

and

$$\widehat{Z}_p \in \{0, 1\}^{M \times \widehat{L}}$$

denote the estimated one-hot membership matrices for users and products, respectively. We

combine them into

$$\hat{Z} = \begin{pmatrix} \hat{Z}_u & 0 \\ 0 & \hat{Z}_p \end{pmatrix} \in \{0, 1\}^{(N+M) \times (\hat{K} + \hat{L})}.$$

Since $\hat{K} = 10$ and $\hat{L} = 11$, the subsequent empirical Bayes analysis contains 21 group-specific distributions of heterogeneous effects. The estimated network memberships therefore determine which users and products are pooled together when learning the group-specific prior distributions, while the observed rating equation is subsequently used to estimate the magnitude of individual user and product effects.

SM.6 Estimated Group Structure in the Amazon User–Product Network

We next examine the latent group structure of the Amazon user–product interaction network. Let $B \in \{0, 1\}^{N \times M}$ denote the bipartite incidence matrix, where $B_{ij} = 1$ if user i rates product j , and $B_{ij} = 0$ otherwise. Group detection is based exclusively on the topology of this interaction network and therefore does not use the numerical rating outcomes. This separation allows the estimated network memberships to be subsequently used in the empirical-Bayes estimation without mechanically constructing the groups from the dependent variable.

To account for the substantial heterogeneity in user activity and product popularity, we apply the degree-corrected bipartite spectral clustering procedure described in Section 3.1. In particular, we construct the degree-normalized biadjacency matrix

$$\tilde{B} = D_u^{-1/2} B D_p^{-1/2},$$

compute its leading singular vectors, and cluster users and products separately using their corresponding spectral representations. The numbers of groups on the two sides are allowed to differ. Comparing alternative candidate specifications using the degree-corrected model-selection criterion yields

$$\hat{K} = 10, \quad \hat{L} = 11,$$

corresponding to ten user groups and eleven product groups.

Figure 6 displays the incidence matrix after reordering users and products according to their estimated group memberships. The visible concentration of links within particular user-group–product-group combinations provides direct evidence of non-uniform matching patterns across the network. Importantly, because the labels of the estimated groups are arbitrary, the strongest interaction blocks need not lie on the main diagonal. For example, user group $U1$ interacts particularly intensively with product group $P4$, while other prominent combinations include $U8$ – $P6$, $U3$ – $P5$, $U7$ – $P8$, and $U10$ – $P10$.

To quantify these differences, Figure 7 reports the density of each estimated user-group–product-group block. For a user group k and product group l , the block density is defined as

$$\hat{\rho}_{kl} = \frac{e_{kl}}{N_k M_l},$$

where e_{kl} denotes the number of observed links between the two groups, and N_k and M_l de-

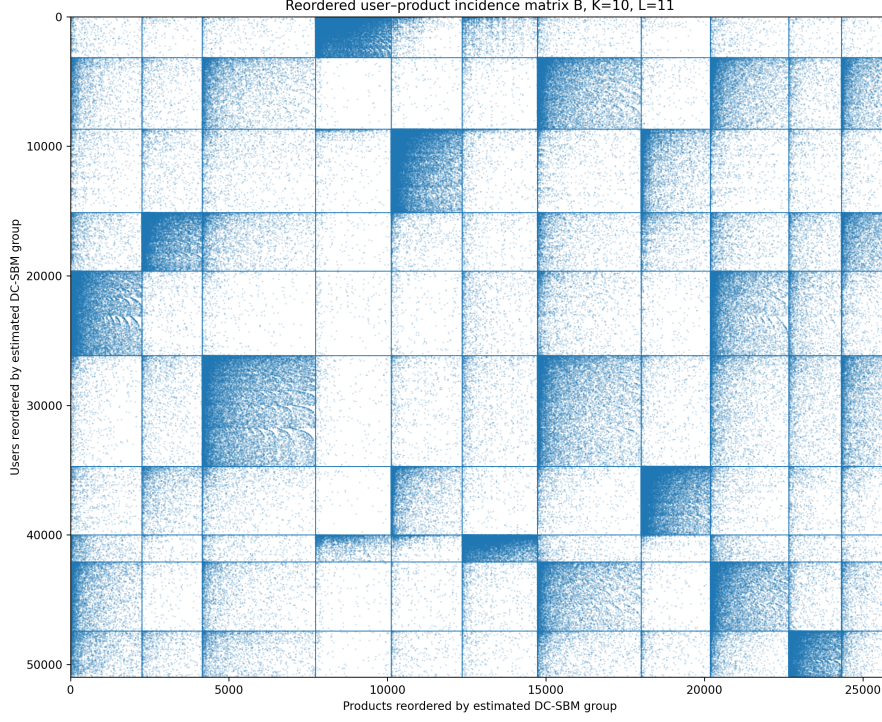


Figure 6: Reordered Amazon user-product incidence matrix. Users and products are reordered according to the estimated degree-corrected bipartite group memberships, with $\widehat{K} = 10$ user groups and $\widehat{L} = 11$ product groups.

note the numbers of users and products in the corresponding groups. The heatmap reveals pronounced variation in interaction intensities across blocks. The largest estimated density occurs between $U1$ and $P4$, with a density of approximately 0.003689, followed by $U8-P6$ with 0.003374, $U3-P5$ with 0.002531, and $U7-P8$ with 0.002527. These patterns suggest that the network contains systematic group-level matching structure beyond differences in individual node degrees.

Table 5 reports the corresponding numbers of observed links in each block. The edge counts display substantial heterogeneity across group pairs. For instance, the $U3-P5$ block contains 36,466 observed interactions, the $U7-P8$ block contains 29,279, and the $U1-P4$ block contains 27,755. The block counts should be interpreted jointly with the density measures, since raw interaction counts also depend on the sizes of the corresponding user and product groups.

Table 6 reports the corresponding block densities. Unlike the raw counts, these statistics normalize for the number of potential user-product links within each group pair and therefore provide a more direct measure of the strength of group-level matching.

Overall, the reordered incidence matrix and block-density estimates provide consistent evidence of pronounced group-level interaction patterns in the Amazon user-product network. The

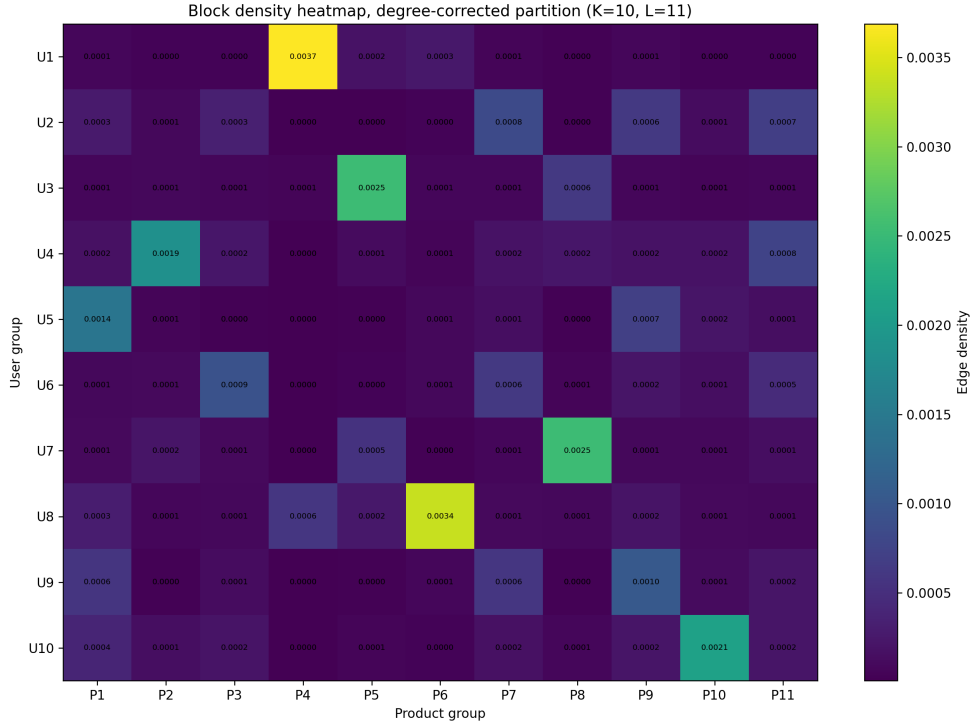


Figure 7: Block-density heatmap for the estimated degree-corrected bipartite partition with $\hat{K} = 10$ user groups and $\hat{L} = 11$ product groups. Each cell reports the observed edge density between the corresponding user and product groups.

Table 5: Observed Edge Counts by Estimated User and Product Groups

	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11
U1	498	215	483	27,755	1,328	1,872	645	183	524	207	192
U2	3,378	879	6,443	136	222	480	15,058	381	8,415	1,213	5,849
U3	1,096	1,022	1,718	979	36,466	1,144	1,524	8,685	1,211	684	738
U4	1,803	15,990	3,647	199	1,215	562	2,697	2,251	2,019	1,370	5,192
U5	21,197	694	588	190	354	1,252	2,922	403	11,442	2,351	1,412
U6	1,539	1,594	28,651	406	939	1,630	17,189	1,107	4,502	2,114	5,995
U7	1,274	2,243	1,863	88	6,507	400	1,325	29,279	1,199	798	1,209
U8	1,433	269	671	2,962	1,159	16,710	720	402	1,013	356	294
U9	6,872	316	2,417	181	370	937	10,230	333	13,603	1,004	1,692
U10	3,093	982	2,493	112	451	361	2,066	607	1,923	12,532	1,159

Notes: Entries report the number of observed user-product links between estimated user group Uk and product group Pl .

Table 6: Edge Densities by Estimated User and Product Groups

	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11
U1	0.000070	0.000036	0.000043	0.003689	0.000188	0.000250	0.000063	0.000027	0.000067	0.000040	0.000040
U2	0.000271	0.000084	0.000326	0.000010	0.000018	0.000037	0.000833	0.000031	0.000615	0.000133	0.000692
U3	0.000076	0.000084	0.000075	0.000064	0.002531	0.000075	0.000072	0.000617	0.000076	0.000064	0.000075
U4	0.000177	0.001861	0.000226	0.000018	0.000120	0.000052	0.000183	0.000228	0.000181	0.000183	0.000753
U5	0.001443	0.000056	0.000025	0.000012	0.000024	0.000081	0.000137	0.000028	0.000709	0.000218	0.000142
U6	0.000080	0.000098	0.000936	0.000020	0.000049	0.000080	0.000614	0.000059	0.000213	0.000149	0.000458
U7	0.000107	0.000223	0.000098	0.000007	0.000548	0.000032	0.000077	0.002527	0.000092	0.000091	0.000149
U8	0.000305	0.000068	0.000090	0.000595	0.000248	0.003374	0.000106	0.000088	0.000196	0.000103	0.000092
U9	0.000574	0.000031	0.000127	0.000014	0.000031	0.000074	0.000589	0.000029	0.001035	0.000114	0.000208
U10	0.000384	0.000144	0.000195	0.000013	0.000056	0.000042	0.000176	0.000077	0.000217	0.002115	0.000212

Notes: For user group k and product group l , the reported quantity is $\hat{\rho}_{kl} = e_{kl}/(N_k M_l)$, where e_{kl} is the observed block-edge count and $N_k M_l$ is the number of possible links within the corresponding block.

estimated structure is neither purely diagonal nor driven by a single high-density block; instead, different user groups display distinct affinities toward different product groups. This finding supports the use of separate latent groups on the user and product sides and provides the first-stage network classification used in the subsequent empirical-Bayes analysis. In particular, the estimated group memberships determine the group-specific prior distributions toward which the individual user and product effects are selectively shrunk.

SM.7 Group-Specific Empirical-Bayes Estimation of User and Product Effects

Building on the network partition described in the preceding subsection, we estimate group-specific distributions of user and product heterogeneity in the Amazon Industrial and Scientific ratings sample. We condition on the supplied partition into $K = 10$ user groups and $L = 11$ product groups throughout this stage. The estimation sample contains $R = 412,947$ rating observations from $N = 50,985$ users on $M = 25,848$ products, identified by `parent_asin`. The mean observed rating is 4.450578. The estimated memberships provide group-specific reference distributions for regularizing individual effects while allowing their locations and dispersions to differ across groups.

For observed rating r , let $i(r)$ and $j(r)$ identify its user and product. We specify the additive model

$$y_r = \beta_0 + \alpha_{i(r)} + \gamma_{j(r)} + \varepsilon_r, \quad r = 1, \dots, R, \quad (56)$$

where α_i and γ_j capture user and product heterogeneity, respectively. The Gaussian working likelihood assumes independent errors with mean zero and common variance σ^2 , independent of the effects conditional on the network and group memberships. Conditional on the estimated memberships $\hat{g}_i$ and $\hat{h}_j$, the effects have independent group-specific priors

$$\alpha_i \mid \hat{g}_i = k \sim \mathcal{N}(\mu_k^s, (\tau_k^s)^2), \quad \gamma_j \mid \hat{h}_j = \ell \sim \mathcal{N}(\mu_\ell^t, (\tau_\ell^t)^2). \quad (57)$$

The model therefore contains 21 group-specific normal distributions. Group labels retain the supplied numbering and have no intrinsic ranking.

We use the preliminary unrestricted two-way fixed-effects (FE) estimates to obtain the intercept and innovation variance. The FE stage imposes the unweighted normalizations $\sum_i \hat{\alpha}_i^{\text{FE}} = 0$ and $\sum_j \hat{\gamma}_j^{\text{FE}} = 0$ and yields $\hat{\beta}_0^{\text{FE}} = 4.425580$. The observed bipartite graph is connected, so the residual degrees of freedom are $R - (N + M - 1) = 336,115$. The corresponding variance estimate is

$$\hat{\sigma}^2 = \frac{\sum_{r=1}^R (y_r - \hat{\beta}_0^{\text{FE}} - \hat{\alpha}_{i(r)}^{\text{FE}} - \hat{\gamma}_{j(r)}^{\text{FE}})^2}{R - (N + M - 1)} = 0.846414. \quad (58)$$

Both quantities are held fixed throughout the subsequent estimation. Displayed values are rounded; the computation uses their stored full precision.

We then implement the plug-in version of the marginal-likelihood method in Section 3.2. Let $\tilde{Y} = Y - \hat{\beta}_0^{\text{FE}} \mathbf{1}_R$, and let B be the $R \times (N + M)$ rating-level incidence matrix, with one user

entry and one product entry equal to one in each row. Stack the effects as $\theta = (\alpha', \gamma')'$, and let $\hat{Z}$ map nodes to their estimated groups. Writing the group means and variances as μ_g and τ_g^2 , define $D = \text{diag}(\hat{Z}\tau_g^2)$. Integrating out the node effects gives the working marginal model

$$\tilde{Y} \mid B, \hat{Z} \sim \mathcal{N}(B\hat{Z}\mu_g, \Omega), \quad \Omega = BDB' + \hat{\sigma}^2 I_R. \quad (59)$$

The group means require a location normalization even though the intercept is fixed: adding a common constant to all user-group means and subtracting it from all product-group means leaves $B\hat{Z}\mu_g$ unchanged. Following the paper, we impose

$$\sum_{\ell=1}^L \frac{M_\ell}{M} \mu_\ell^t = 0, \quad M_\ell = \sum_{j=1}^M \mathbf{1}\{\hat{h}_j = \ell\}. \quad (60)$$

Thus, the 21 reported group means contain 20 free location parameters. No additional mean-zero restriction is imposed on the user-group means.

The group parameters are estimated jointly by maximizing

$$\begin{aligned} \ell(\mu_g, \tau_g^2) = & -\frac{R}{2} \log(2\pi) - \frac{1}{2} \log |\Omega| \\ & - \frac{1}{2} (\tilde{Y} - B\hat{Z}\mu_g)' \Omega^{-1} (\tilde{Y} - B\hat{Z}\mu_g), \end{aligned} \quad (61)$$

subject to equation (60) and nonnegative group variances. For each variance vector, the group means are obtained by constrained generalized least squares. The resulting profile likelihood is optimized over the 21 group variances. This is marginal maximum likelihood with fixed plug-in nuisance parameters, rather than restricted maximum likelihood or joint estimation of the innovation variance.

The criterion uses the original ratings and the full covariance induced by shared users and products. It therefore incorporates the network dependence that is omitted when preliminary FE estimates are treated as independent signals with sampling-variance proxies $\hat{\sigma}^2/d_i$. Computation uses exact determinant and Woodbury identities, followed by elimination of the diagonal user block and Cholesky factorization of the product-side Schur complement. This avoids forming the $R \times R$ observation covariance matrix. Analytic variance scores are evaluated using exact inverse-diagonal terms.

Model-level results. Table 7 summarizes the sample, fixed inputs, and convergence diagnostics. The marginal-likelihood optimization converges in eight iterations using nine likelihood evaluations. The largest absolute score with respect to the log group variances is 8.995×10^{-6} , and all estimated variances are strictly positive. The likelihood and derivative calculations were checked against direct covariance-matrix calculations on a small test model. These checks and the reported scores establish numerical consistency and convergence of the implementation; they do not constitute a general proof of global optimality for variance-component likelihoods.

Table 7: Sample characteristics, fixed inputs, and marginal-likelihood diagnostics.

Statistic	Value
Rating observations	412,947
Users	50,985
Products	25,848
User groups	10
Product groups	11
Connected components	1
Mean observed rating	4.450578
Fixed FE intercept	4.425580
Fixed innovation variance (df corrected)	0.846414
FE residual degrees of freedom	336,115
Free group-mean parameters	20
Group-variance parameters	21
Marginal log likelihood	-586,357.855799
Optimization iterations	8
Likelihood evaluations	9
Largest absolute log-variance score	8.995×10^{-6}
Largest absolute constrained mean score	1.689×10^{-12}

Notes: The intercept and innovation variance are fixed at their preliminary FE estimates. The marginal log likelihood includes the Gaussian constant. Scores are derivatives of the likelihood criterion, rather than changes in individual effects between iterations. Values are rounded for presentation.

Estimated group-specific heterogeneity. Tables 8 and 9 report the estimated group-specific prior distributions. User prior means range from -0.170941 in U3 to 0.144425 in U1, a difference of 0.315367 rating points. Holding product heterogeneity constant, these estimates indicate systematic differences in users' rating tendencies. The network partition therefore contains information about rating behavior even though the groups are estimated from the interaction structure rather than from rating outcomes.

Estimated within-group user variances also differ substantially. U1 has the smallest variance, 0.024957 , corresponding to a prior standard deviation of 0.157978 , whereas U3 has the largest variance, 0.478668 , corresponding to a standard deviation of 0.691859 . U7 also exhibits relatively high dispersion, with prior variance 0.351302 . Thus, a relatively favorable group-level

rating tendency can coexist with comparatively homogeneous user effects, while a group with a negative mean can exhibit considerable individual variation. Allowing both prior locations and dispersions to vary makes the targets and strength of regularization group specific.

Table 8: Estimated user-group prior parameters from plug-in marginal maximum likelihood.

Group	Nodes	$\hat{\mu}$	$\hat{\tau}^2$	$\hat{\tau}$
U1	3,152	0.144425	0.024957	0.157978
U2	5,531	0.060736	0.165517	0.406838
U3	6,429	-0.170941	0.478668	0.691859
U4	4,515	0.035314	0.205083	0.452861
U5	6,525	0.015160	0.242132	0.492069
U6	8,558	0.062127	0.171099	0.413641
U7	5,295	-0.074558	0.351302	0.592708
U8	2,085	0.099826	0.088797	0.297988
U9	5,314	0.032102	0.203814	0.451458
U10	3,581	0.011992	0.225873	0.475261

Notes: The columns report group sizes and estimated prior means, variances, and standard deviations. Prior SD measures within-group heterogeneity, not the standard error of the estimated prior mean. Means follow the normalization in equation (60). Values are rounded to six decimal places.

Product prior means range from -0.178940 in P8 to 0.120663 in P4, a difference of 0.299603 rating points. P5 also has a low prior mean, -0.175566 . Conditional on user-specific rating tendencies, products in P4 therefore have a higher average product-specific rating component than products in P5 and P8. The product groups inferred from the network are informative about systematic variation in product evaluations.

Product prior variances show substantial differences in within-group dispersion. P4 has the smallest variance, 0.008926 , corresponding to a prior standard deviation of 0.094478 . P8 and P5 have the largest variances, 0.149291 and 0.145862 , with standard deviations of 0.386382 and 0.381919 , respectively. Product effects in P4 are therefore much more concentrated around their group mean. The larger variances in P5 and P8 allow substantial product-level heterogeneity within these groups. These effects describe conditional rating components and should not be interpreted as causal measures of product quality or user behavior.

Comparison with the FE-based approximation. To compare prior means across implementations, we first express both sets of estimates under the normalization in equation (60). This requires adding 0.006094 to the previously reported user-group means and subtracting the same constant from the product-group means; the full-precision adjustment leaves all user-product mean sums unchanged. After alignment, the largest absolute difference between corresponding group means is 0.019819 rating points. The estimated prior locations are therefore numerically

Table 9: Estimated product-group prior parameters from plug-in marginal maximum likelihood.

Group	Nodes	$\hat{\mu}$	$\hat{\tau}^2$	$\hat{\tau}$
P1	2,252	-0.021073	0.070295	0.265132
P2	1,903	0.006960	0.059220	0.243352
P3	3,576	0.064244	0.047835	0.218712
P4	2,387	0.120663	0.008926	0.094478
P5	2,241	-0.175566	0.145862	0.381919
P6	2,375	-0.011936	0.050207	0.224070
P7	3,270	0.073730	0.039899	0.199747
P8	2,188	-0.178940	0.149291	0.386382
P9	2,473	0.008051	0.063521	0.252033
P10	1,655	0.016329	0.068365	0.261468
P11	1,528	0.027312	0.054498	0.233448

Notes: The columns report group sizes and estimated prior means, variances, and standard deviations. Prior SD measures within-group heterogeneity, not the standard error of the estimated prior mean. Means follow the normalization in equation (60). Values are rounded to six decimal places.

close across the two methods.

The variance estimates differ more noticeably. All 21 variances are lower under the full marginal likelihood. User-group variances decline by 1.33%–7.63%, with a median reduction of 3.11%. Product-group variances decline by 5.65%–40.01%, with a median reduction of 13.51%. The largest proportional change occurs in P4, whose estimated variance decreases from 0.014880 to 0.008926. The approximation thus delivers similar prior locations in this sample, but it is not numerically equivalent to the full marginal-likelihood method, particularly for product-side dispersion.

Empirical-Bayes shrinkage under the fitted priors. Given the estimated group parameters, let $\hat{D} = \text{diag}(\hat{Z}\hat{\tau}_g^2)$. The associated empirical-Bayes posterior mean is

$$\hat{\theta}^{\text{EB}} = \hat{Z}\hat{\mu}_g + \hat{D}B'(B\hat{D}B' + \hat{\sigma}^2 I_R)^{-1}(\tilde{Y} - B\hat{Z}\hat{\mu}_g). \quad (62)$$

The intercept remains fixed at $\hat{\beta}_0^{\text{FE}}$. Equivalently, the posterior mean minimizes

$$Q(\alpha, \gamma) = \frac{1}{2\hat{\sigma}^2} \sum_{r=1}^R (y_r - \hat{\beta}_0^{\text{FE}} - \alpha_{i(r)} - \gamma_{j(r)})^2 + \frac{1}{2} \sum_{i=1}^N \frac{(\alpha_i - \hat{\mu}_{g_i}^s)^2}{(\hat{\tau}_{g_i}^s)^2} + \frac{1}{2} \sum_{j=1}^M \frac{(\gamma_j - \hat{\mu}_{h_j}^t)^2}{(\hat{\tau}_{h_j}^t)^2}. \quad (63)$$

The Gaussian posterior mean and mode coincide conditional on these plug-in inputs.

The conditional update for user i can be written as

$$\begin{aligned}\hat{\alpha}_i^{\text{EB}} &= w_i \left[\frac{1}{d_i} \sum_{r:i(r)=i} \left(y_r - \hat{\beta}_0^{\text{FE}} - \hat{\gamma}_{j(r)}^{\text{EB}} \right) \right] + (1 - w_i) \hat{\mu}_{g_i}^s, \\ w_i &= \frac{(\hat{\tau}_{g_i}^s)^2}{(\hat{\tau}_{g_i}^s)^2 + \hat{\sigma}^2/d_i},\end{aligned}\tag{64}$$

where d_i is the user's number of ratings. An analogous expression applies to products. For a fixed rating count, a smaller prior variance places greater weight on the group mean. For a fixed prior variance, more ratings place greater weight on the individual's adjusted outcomes. Because those outcomes depend on the jointly determined effects on the opposite side, the posterior estimator generally cannot be expressed as a scalar weighted average of a preliminary FE estimate and its group mean. The scalar weights characterize conditional updates; they are not substituted for the network covariance in hyperparameter estimation.

Uncertainty and interpretation. Conditional on the fixed intercept, innovation variance, memberships, and estimated hyperparameters, the full Gaussian posterior covariance is

$$\hat{V} = \left(\frac{B'B}{\hat{\sigma}^2} + \hat{D}^{-1} \right)^{-1}.\tag{65}$$

Its diagonal elements account for posterior dependence propagated through the network. In contrast, the scalar expression $(d_v/\hat{\sigma}^2 + 1/\hat{\tau}_{q(v)}^2)^{-1}$ is the conditional variance of node v when the other node effects are held fixed. Neither measure incorporates estimation uncertainty in the intercept, innovation variance, memberships, or prior parameters.

Overall, the fitted marginal model reveals substantial heterogeneity in both the locations and dispersions of the group-specific priors. The resulting shrinkage targets differ across network-defined groups, and the estimated dispersion parameters allow the strength of pooling to vary across groups and between users and products. These findings characterize the fitted distribution of heterogeneity. Assessing gains in predictive performance or effect-estimation accuracy requires additional evaluation beyond the reported marginal-likelihood results.

SM.8 Predictive Value of Network-Informed Empirical Bayes

Motivation and objective. In sparse two-way data, the fixed effect for an individual may be estimated imprecisely because that individual is observed in relatively few matches. Empirical Bayes (EB) estimation addresses this problem by shrinking noisy fixed-effect estimates toward an estimated prior mean. The amount and direction of shrinkage depend on the prior distribution. A pooled EB procedure assigns a common prior distribution to all users and another common prior distribution to all products. If users or products with different effect distributions have different matching patterns, this pooling scheme may combine dissimilar individuals.

This application examines whether the bipartite matching network contains useful information for choosing which users and products should share a prior distribution. The empirical question is whether co-spectral clustering of the training network, followed by group-specific EB shrinkage, improves prediction of held-out ratings relative to pooled EB and other benchmarks. We also compare EB estimators with two-way fixed effects to assess the predictive value of shrinkage more generally. The design is predictive: it evaluates out-of-sample rating prediction and does not, by itself, establish recovery of the latent effect distributions or causal effects.

Data and prediction model. The analysis uses the Industrial and Scientific ratings data. The usable sample contains 412,947 user–product ratings, covering 50,985 users and 25,848 products between 2001 and 2023. Each observed rating corresponds to a distinct user–product pair. For user i , product j , and rating Y_{ij} , the baseline two-way additive model is

$$Y_{ij} = \alpha_i + \gamma_j + \varepsilon_{ij},$$

where α_i is a user effect, γ_j is a product effect, and ε_{ij} is an idiosyncratic error. All estimates and network groups in a given split are constructed using training data only.

For the network-based EB estimator, let g_i and h_j denote the user and product groups, respectively. The group-specific prior distributions are

$$\alpha_i \mid g_i \sim \mathcal{N}\left(\mu_{g_i}^u, (\tau_{g_i}^u)^2\right), \quad \gamma_j \mid h_j \sim \mathcal{N}\left(\mu_{h_j}^p, (\tau_{h_j}^p)^2\right).$$

The network used for clustering is the bipartite graph induced by training observations: a user–product edge is present when their rating is included in the training sample. Co-spectral clustering is applied to this training graph to estimate the user and product memberships. The number

of groups is selected using the CBIC procedure described in the appendix, over the candidate counts specified for the user and product sides.

The prior parameters are estimated by fitting heteroskedastic normal models to the two-way fixed-effect estimates. The measurement-variance proxy for an individual effect is the estimated training disturbance variance divided by that individual’s number of training ratings. The posterior means are then computed jointly under the additive user–product model, conditional on these plug-in prior parameters. Accordingly, the results below evaluate the implemented plug-in EB procedure. They should not be interpreted as a direct evaluation of an estimator that maximizes the exact marginal likelihood of the original rating model.

Evaluation design. The analysis uses two complementary holdout schemes. First, temporal evaluation uses test years 2021, 2022, and 2023. Models are estimated using ratings available before the relevant test period; the preceding year is used for validation. This design assesses prediction of future ratings while respecting the time ordering of the data. Second, random edge evaluation holds out 20 percent of user–product ratings and repeats the random split three times. This design assesses prediction for randomly held-out links when the remaining links are available for training.

The primary evaluation sample in each split consists of warm ratings: both the user and the product must appear in the training data. The temporal test folds contain 83,636 warm ratings in total, out of 149,892 ratings in the corresponding test periods. Thus, the temporal results apply to approximately 56 percent of those test ratings; ratings involving a user or product absent from training are not included in the warm-node prediction scores. By comparison, almost all randomly held-out ratings are warm. The two designs therefore address different prediction tasks, and the random-edge results should not be taken as evidence about cold-start prediction.

For each split, we compare the following estimators:

1. two-way fixed effects without EB shrinkage;
2. pooled EB, with one prior distribution for users and one for products;
3. degree-group EB, which forms groups using training-network degree bins;
4. degree-matched permutation EB, a placebo benchmark based on permuted group assignments matched to degree; and

5. network-group EB, which uses the co-spectral groups estimated from the training bipartite graph.

We also examine selected alternative network group counts as a sensitivity analysis. The primary predictive metrics are the root mean squared error (RMSE) and mean absolute error (MAE), computed over the eligible held-out ratings. Standard errors for paired predictive comparisons are clustered by user and product.

Predictive results. Table 10 reports aggregate RMSE by evaluation design. EB shrinkage improves predictive accuracy relative to the unshrunk two-way fixed-effects estimator in both designs. In the temporal evaluation, pooled EB reduces RMSE from 1.322 to 1.157, a reduction of approximately 12 percent. In the random-edge evaluation, pooled EB reduces RMSE from 1.067 to 0.989, a reduction of approximately 7 percent. Network-group EB also improves on two-way fixed effects, with RMSEs of 1.159 and 0.991 in the temporal and random-edge evaluations, respectively. These results indicate that shrinkage is useful for predicting ratings in this sample.

The results do not, however, show an incremental predictive gain from using network-estimated groups. In the temporal evaluation, network-group EB has an RMSE of 1.159, compared with 1.157 for pooled EB and 1.158 for degree-group EB. In the random-edge evaluation, its RMSE is 0.991, compared with 0.989 for pooled EB and 0.989 for degree-group EB. The network-group estimator is therefore slightly less accurate than the simpler EB benchmarks in these aggregate comparisons. The degree-matched permutation benchmark also performs similarly to pooled EB, which provides no evidence here that the estimated network labels add predictive information beyond the information captured by the simpler benchmarks.

Table 10: Held-Out Rating Prediction

Evaluation design	Two-way FE	Pooled EB	Degree-group EB	Network-group EB
Temporal test	1.322	1.157	1.158	1.159
Random edge holdout	1.067	0.989	0.989	0.991

Notes: Entries are RMSEs over held-out ratings for which both endpoints are observed in the training data. The temporal results pool the 2021–2023 test folds. The random-edge results pool the three 20-percent edge holdouts. Lower values indicate better predictive performance.

The paired comparisons reinforce this interpretation. In each of the three random-edge splits, the two-way-clustered confidence interval for the difference between pooled EB and network-group EB favors pooled EB. For the 2023 temporal test fold, the corresponding interval includes zero, so that fold does not establish a difference between the two methods. The estimated

network-group disadvantage is small in magnitude, even where the paired comparison is statistically distinguishable from zero. The evidence should therefore be read as showing no demonstrated predictive advantage from network-based grouping in this application, rather than as establishing that network grouping is uniformly harmful.

The CBIC-selected group counts also vary across training samples. The procedure selects $(K, L) = (1, 1)$ in some temporal folds, $(6, 6)$ or $(6, 7)$ in others, and $(3, 4)$ in each random-edge split. This variation indicates that the selected group structure depends on the training window and holdout design. It also helps explain why the results do not support a stable predictive improvement from group specificity. The alternative group-count specifications produce similarly small predictive differences, rather than a consistent improvement over pooled EB.

Product rankings and interpretation. As a secondary exercise, we compare estimated product effects with adjusted test-period ratings and examine the stability of product rankings across temporal folds. EB rankings are generally more stable across folds than rankings based on two-way fixed effects. However, pooled EB rankings are at least as stable as network-group EB rankings, and their average association with adjusted test ratings is slightly higher in the temporal evaluation. The rank-validation exercise is suggestive rather than conclusive: the set of products meeting the minimum test-rating threshold varies across estimators, and adjusted test-period averages are noisy measures of underlying product effects. We therefore do not interpret these ranking comparisons as evidence that any estimator recovers true product quality.

Limitations. Several qualifications are important for interpreting the findings. First, the predictive analysis evaluates warm-node ratings and does not measure performance for new users or new products. Second, the hyperparameter procedure uses a two-way fixed-effects approximation with a degree-based measurement-variance proxy. The reported comparison therefore concerns this implemented plug-in estimator, and further analysis using the paper’s exact marginal-likelihood procedure would be useful. Third, the training bipartite graphs have multiple connected components in some temporal folds. In such samples, the rating likelihood does not identify relative fixed-effect locations across components without additional normalization or prior restrictions. This feature can affect comparisons involving unshrunk two-way fixed effects and should be considered when interpreting the temporal results.

Overall, the application provides evidence that EB shrinkage improves held-out rating pre-

diction relative to unshrunk two-way fixed effects. It does not provide evidence that co-spectral groups improve prediction beyond pooled or degree-based EB in this dataset. The network-based estimator remains useful for studying whether matching patterns reveal interpretable heterogeneity in effect distributions, but any such descriptive or structural contribution should be distinguished from the predictive conclusion of this exercise.